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High-frequency dynamics of heterogeneous slender structures

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ABSTRACT

This paper gives an overview of the theoretical modeling of high-frequency linear dynamics of built-up structures including the influence of uncertainties by a probabilistic approach. Its analytical developments are enlightened by a preliminary discussion on the vibrational responses of such systems as observed from some experiments conducted in a broad frequency range of excitation. The paper first reviews the main engineering approaches used so far to address the higher frequency domain, namely the statistical energy analysis and the vibrational conductivity analogy. Both methods establish heuristic steady diffusion equations to describe the spatial distribution of the vibrational energy. It is then argued that several limitations and assumptions which restrict their range of validity may be released if a wave transport model is invoked. The latter describes the multiple reflections of high-frequency elastic waves in heterogeneous (possibly random) media adopting a kinetic point of view pertaining to the associated energy density. Transient transport equations evolve into unsteady diffusion equations after long times, supporting in this respect the engineering approaches. Thus the second part of the paper is devoted to a generic presentation of some recent works on kinetic transport models for application to structural dynamics. This objective requires the extension of the existing results of that theory to include dissipation and boundary effects. The proposed models are illustrated by a numerical example showing their consistency with an SEA computation, and the concurrence of a time domain simulation with a frequency domain result.

1. Overview of the paper

The dynamic response of built-up engineering structures to low-frequency (LF) excitations can be predicted efficiently by reduced models derived from modal analyses. Throughout the paper the terminology "built-up structure" refers to a mechanical system constituted by the assembly of (i) several more simple subsystems, such as beams, plates, cylindrical shells, etc., or (ii) secondary equipments attached to a main structure, such as electromechanical or hydraulic devices. In the former case, the substructures may have high stiffness contrasts or exhibit a repetitive, indeed periodic pattern. This situation is however not considered in this paper because it is anticipated that actual structures would never be perfectly periodic even if they are designed to be so. Built-up systems such as aircraft fuselages, ship hulls, or car bodies typically show such features. Modal models, when eventually updated by appropriate methods [1], compare usually very well with experimental data at low frequencies. However, they may deteriorate rapidly when the frequency is increased, mainly because of the increased influence of the irreducible uncertainties related to the higher-order natural modes and

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frequencies [2], which contribute to lower the quality of the computed modal reduction bases. So built-up structures have very different dynamic responses at higher frequencies (HF) as compared to the ones observed in the low-frequency modal range [3,4]; this issue is discussed in greater detail in Section 2. Alternative modeling strategies have thus to be developed for the simulation and prediction of HF responses. In particular they should take into account the various sources of uncertainties. *Wave transport models* provide a relevant description of HF vibrations and wave propagation phenomena in heterogeneous media, as we try to demonstrate it subsequently. The aforementioned heterogeneous media are typically not known precisely and are thus modeled as realisations of random media with known statistics, *e.g.* unknown arbitrary samples among a parent population of presumed identical structures. The derivation and tentative validation of these models is addressed in Sections 3 and 4, which constitute the main contribution of the paper. The motivations for these developments are clarified below.

Transport and diffusion in built-up structures: As seen from the experiments expounded in [3,4], engineering structures exhibit a typical diffusive behaviour in their higher frequency ranges of vibration. Here the qualification "high" corresponds to frequency bands extending to many times the fundamental natural frequencies of the systems in consideration. Such vibrational responses can be estimated by the *statistical energy analysis* (SEA, see [2,5–13] or [14, Chapter V, Section 8; 15, Chapter 7] for short introductions) or the *vibrational conductivity analogy* (VCA, see [16–24]) for steady or unsteady excitations of which spectra extend to the HF ranges. The loads may be either deterministic or random, such as turbulent boundary layers, external pressure fields, or track and road profiles. The quantities computed in SEA are the averaged (in a sense precised in Section 2.3) vibrational energies integrated over subsystems in a built-up structure. This lack of spatial resolution has prompted the development of local energetic approaches, among which VCA has emerged as a possible alternative to SEA. SEA and VCA both predict a diffuse vibrational state of the components of the system, relying however on some crucial assumptions in order to effectively obtain diffusion. These methods are reviewed in Section 2 with a particular focus on their range of validity in this respect.

The main argument elaborated subsequently in Section 3 is that diffusion models can rather be derived from wave transport models. Indeed the HF responses in [3,4] may be described by linear transport equations for the *energy density* associated to the strongly oscillating (HF waves) solutions of the elastic wave equation modeling transient structural dynamics. More generally this result holds for all classical and quantum wave systems, including acoustic or electromagnetic waves [25–34]. It has been specialised to slender viscoelastic structures, typically beams, plates and shells, and fluid-saturated poro-viscoelastic media in [35–38] for applications to built-up systems. The mathematical model is derived from the semiclassical analysis of HF solutions of wave equations. It also allows to track the energy paths (rays) within the propagation medium, thus describing all energy fluxes in slender structures. For these reasons the underlying theory is believed to provide a rational framework for the validation, and generalisation, of SEA and VCA. Although it has received a considerable attention in the last decades in the physics literature [39–48], this approach is however less developed in the structural–acoustics literature [49–54]. In addition, it is applicable in the transient domain, contrary to SEA or VCA which essentially focus on the frequency domain. This refinement is needed in order to capture the reflection and scattering phenomena of HF vibrational waves, as they traverse and re-traverse a structure or a component many times. Indeed, by repeatedly encountering changes of geometry and dynamic properties of their supporting medium, waves are partially reflected, transmitted or diffracted leading to a redistribution of the incident energy into many directions. The strength of reflection and scattering increases as the frequency increases, or as the wavelength decreases. These phenomena ultimately yield the aforementioned diffusive regime once the incident wave energy has been spread rather uniformly over its whole support. Transport models are constructed adopting a *kinetic* point of view of wave propagation, by which waves are described in the *phase space* position $\times$ wave vector. The significance of using such a description is discussed next.

Kinetic models for structural dynamics: The relevance of a local wave approach to the analysis of HF vibrations of built-up systems has already been recognised for a long time in the structural–acoustics literature; see for example Refs. [49,55–60]. In these works the wave fields are often resolved into a direct, ballistic field and a scattered, reverberant or diffuse field. This idealisation is not necessary in the proposed kinetic models of wave propagation, since they encompass both the direct and scattered fields in a unique description. The natural tool to derive kinetic equations for classical or quantum waves is the Wigner transform of two such wave fields [61]. It is adapted to the characterisation of multiply scattered wave fields propagating in many directions at each point in the physical space since it is a distribution in phase space. The Wigner transform of finite-energy wave fields converges to a positive Hermitian measure, called the Wigner measure, as their wavelength decreases (HF limit). The Wigner measure is related to the energy density of the waves in this very limit (under suitable hypotheses [26, Proposition 1.7]). It is also shown to satisfy a Liouville transport equation in a slowly varying medium with respect to the small wavelength, or a radiative transfer equation if the correlation length of the heterogeneities is comparable to the wavelength [27–33], or a Fokker–Planck diffusion equation if some heterogeneities have correlation length much larger than the wavelength [33,62]. All of them ultimately evolve into classical spatial diffusion equations of heat conduction type at large times and/or propagation distances [28,33,63–68]; these asymptotics are recognised as the diffusion limit invoked in SEA and VCA.

A close perspective is given by the geometrical optics approach of HF wave propagation phenomena; see e.g. [69–73]. In fact ray methods are the oldest and best known techniques in this field and have been used in structural dynamics for a long time [74–81]. Replacing the solution of a wave equation by its ray approximation yields an eikonal and a transport equation for the phase and the amplitudes at increasing orders, respectively. Re-interpreting it in a Lagrangian kinematical description, the latter is a Liouville equation in phase space. Hence the semiclassical analysis of wave systems has deep connections with geometrical optics and can be understood in this more familiar framework. The Wigner measure may be seen as a phase space density of energy "particles" in this setting, such that a kinetic equation is a natural candidate to follow its evolution in an heterogeneous medium. This point of view is elaborated further on in Section 3.

Now boundary conditions and dissipation effects have not been considered so far, though they are essential for the description of the dynamic response of a built-up structure. The kinetic models outlined above have to be supplemented by *ad hoc* boundary and/or interface conditions in order to define a well-posed boundary value problem [63,64]. The derivation of boundary and interface conditions for energetic (quadratic) quantities consistent with the boundary and interface conditions imposed to the underlying wave fields has been addressed in [82–86]. They may be written as power flow reflection/transmission operators for the energy rays—along the same lines as in geometrical optics [72]. Specular-like transverse boundary reflections, diffuse reflections, or fluid–structure coupling for example may be treated as a particular case. A formal derivation of these operators at interfaces between slender substructures such as beams or shells in the HF limit has been proposed in [38,87]. More recently, a systematic derivation of the elastic energy density boundary conditions has been developed in [34]. The analysis however does not account for the glancing set (at critical incidence) and the transport of energy along interfaces by the so-called gliding rays. In fact the occurrence of critical incidences, diffraction phenomena and polarisation conversions as in classical elastic wave propagation [88] raises some serious theoretical difficulties yet partially unsolved. Regarding dissipation phenomena, they can be accounted for with a viscosity coefficient (or a rate of relaxation) stemming from a memoryless Kelvin–Voigt constitutive model [89,90], since it may be shown that memory effects in viscoelastic materials have no influence on the HF transport regimes [36,91]. Then kinetic models in bounded, dissipative media with general transverse or diffuse boundary conditions for the energy fluxes may be derived from these results [92,93]. They are detailed in Section 4, together with some elements for the numerical resolution of transport equations. Here a numerical example is also presented, dealing with an assembly of thick shells. The emergence of a diffusive regime at late times is demonstrated, and it is argued that the latter is precisely the one invoked in SEA and VCA to derive steady-state global or local diffusion equations by heuristic arguments. This example thus enlightens the consistency of the proposed theory with the engineering approaches.

Outline of the paper: In short, Section 2 below reviews some experimental data dealing with HF vibrations of built-up structures and summarises the usual SEA and VCA approaches for interpreting them. Its purpose is to introduce and motivate the theoretical developments presented in the subsequent sections: kinetic equations for elastic wave propagation in heterogeneous media (Section 3), and extensions of kinetic models for their application to structural dynamics, as well as a brief overview of some dedicated numerical methods with an example (Section 4). Conclusions and perspectives of future researches are offered in Section 5.

2. Existing energy approaches for mid- and high-frequency vibrations

In this section, the distinguishing features of mid- to high-frequency structural dynamic responses are first introduced qualitatively and quantitatively, as well as some clarifications of the terminology used in the remaining of the paper. It continues with a presentation of the main analytical models used by engineers to predict the structural vibrations in the HF ranges, namely the statistical energy analysis (SEA) in Section 2.3 and the vibrational conductivity analogy (VCA) in Section 2.4. The links between the former global approach and the latter local approach are established in the frame of the classical setting of three-dimensional elastodynamics. The merits and weaknesses of these methods are discussed as well, in order to introduce the research strategy described in Section 2.5. The main conclusions obtained from this discussion are considered as keys for understanding the motivations and objectives of the different results presented in the subsequent sections of this paper. Thus the following introductory presentation does not pretend to be exhaustive; its purpose is rather to initiate the approach developed in Sections 3 and 4 from the observations summarised here.

2.1. Frequency response function of built-up systems

The vibrational responses observed for a large-scale experimental model in [3] provide a good illustration of the main concerns raised by the higher frequency ranges. Fig. 1 shows an overall CAD view of this structure and a sketch of the locations of the excitations as well as the four sets of accelerometers spread on it. The structure is divided along its longitudinal axis (denoted by y on these sketches) into nine segments separated by vertical bulkheads constituted by non-stiffened plates. The segments were assigned different lengths to break the periodicity. White-noise forces are successively applied at four points at one end, say segment #1. To eliminate the contribution of internal acoustics, foams have been suspended in all cavities. The structure is made from an aluminium alloy, and reciprocity measurements have shown that it globally has a linear behaviour for the entire frequency range considered, namely $I_{\rm exp}=[50-5000]$ Hz. The plate bending wavelength at this maximum frequency is about 5 cm. This example and the various conclusions which have arisen from its detailed examination are expounded in [3], together with a discussion on the numerical simulations performed in the frequency range $I_{\rm num}=[100-1000]$ Hz. As an example of the various measurements done on this structure, Fig. 2 displays the estimated mechanical energies as densities vs. the circular frequency in different segments. Three characteristic frequency domains denoted by "LF", "MF" and "HF" are shown in Fig. 2. In the low-frequency range labelled LF, here for frequencies f up to about 250 Hz, the estimated mechanical energies have the same average levels for the three segments.

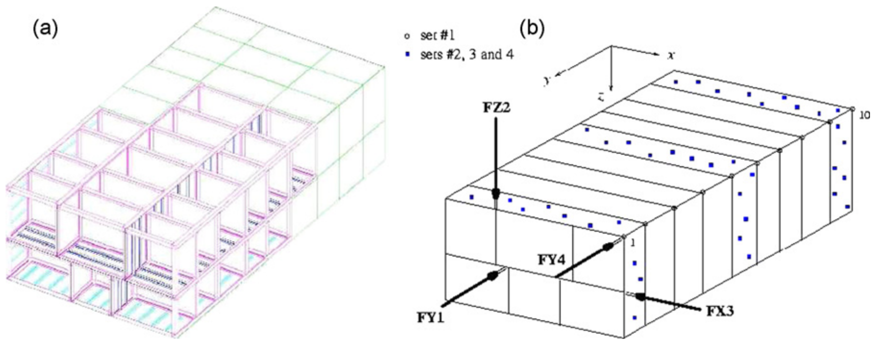

Fig. 1. (a) CAD view of the experimental structure: length = 5.3 m, width = 2.5 m, height = 1.4 m, and plate thickness = 1.2 mm. (b) Location of the excitations and accelerometers. After [3].

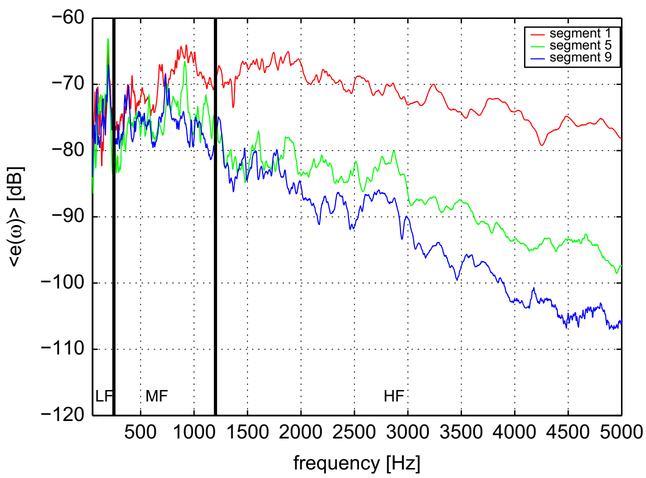

Fig. 2. Estimated mechanical energy densities of the experimental structure of Fig. 1 for segments #1 (at the end of the structure where the loads are applied), #5 (intermediate) and #9 (the other end) considering the excitation "FX3" in the frequency range I_{exp}. These estimates are computed as the mass-weighted mean square velocities of the plates constituting each segment as measured by the randomly distributed accelerometers and $dB_{ref} = 10 \times \log_{10}(1\ \text{kg m}^2/\text{s}^2)$. After [3].

Thus for such frequencies the vibratory energy propagates broadly to the entire structure, and does not remain localised near the excitation. On the contrary in the high-frequency range labelled HF ($f \gtrsim 1200$ Hz), the energy levels decrease significantly when observed at increased distances from the excitation. They are also steadily decreasing with the frequency. Thus for such frequencies the vibratory energy remains localised close to the excitation and diffuses only weakly to the other parts of the structure. For the intermediate frequencies, the so-called mid-frequency range labelled MF, the general trend is that the energy levels in segments #5 and #9 are comparable but lower than the one in segment #1. Thus the vibratory energy gets only partially localised close to the excitation, the remaining being distributed in the entire structure.

The differences between these different behaviours are also particularly appealing on the experimental results described in [4]. Here the authors consider the surface and interior response of a Cessna Citation fuselage section (see Fig. 3) under different forcing functions evaluated through spatially dense scanning measurements. The experiment was carried out in a laboratory environment in the frequency range $I_{exp} = [10-1000]$ Hz, for which the minimum bending wavelength of the aluminium fuselage shell is about 9 cm. Contrary to the previous example, the contribution of internal acoustics has been fully accounted for in the measurements and numerical simulations. Fig. 4 displays the recorded phase of the normal surface velocity field for a point force applied at a rib/stringer intersection at one end of the fuselage. Fig. 5 displays the magnitude of the normal surface velocity field for the same point force excitation. It compares the measured and numerically predicted distributions, using a finite/infinite element model accounting for both internal and external acoustic media. The other excitations considered in this study, namely a point force applied to a flexible thin walled panel area between stiffeners at one end of the fuselage, and external acoustic source, exhibit however a similar categorisation as underlined by the authors. Fig. 5 shows marked modal localisation and clustering effects, whereas the phase patterns in Fig. 4 as the frequency increases are clear manifestations of the aforementioned energy transfer phenomena. Indeed, energy flows in an elastic medium are proportional, in a first approximation, to the *gradient* of the phase of the velocity field. At low frequencies, all the energies are concentrated on the global eigenmodes and no spatial transfer occurs because

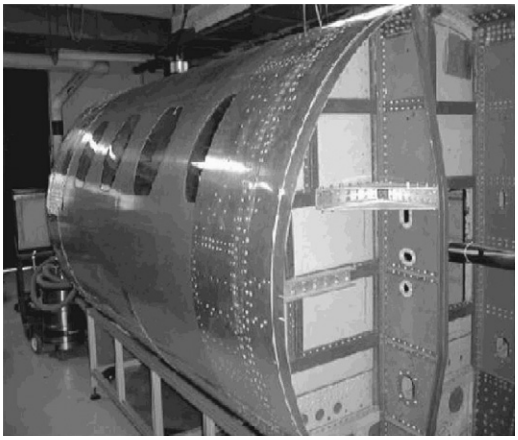

Fig. 3. Cessna citation fuselage section and support structure: interior radius = 0.81 m, length = 2.55 m. After [4].

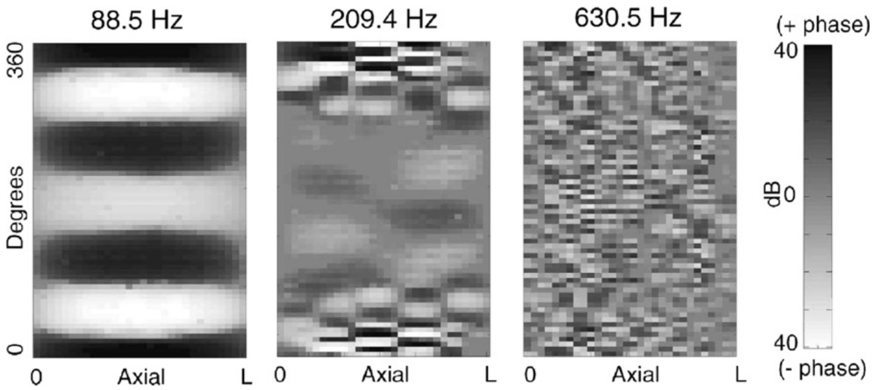

Fig. 4. Measured phase of the normal surface velocity in circumferential angle vs. axial position for different frequencies. After [4].

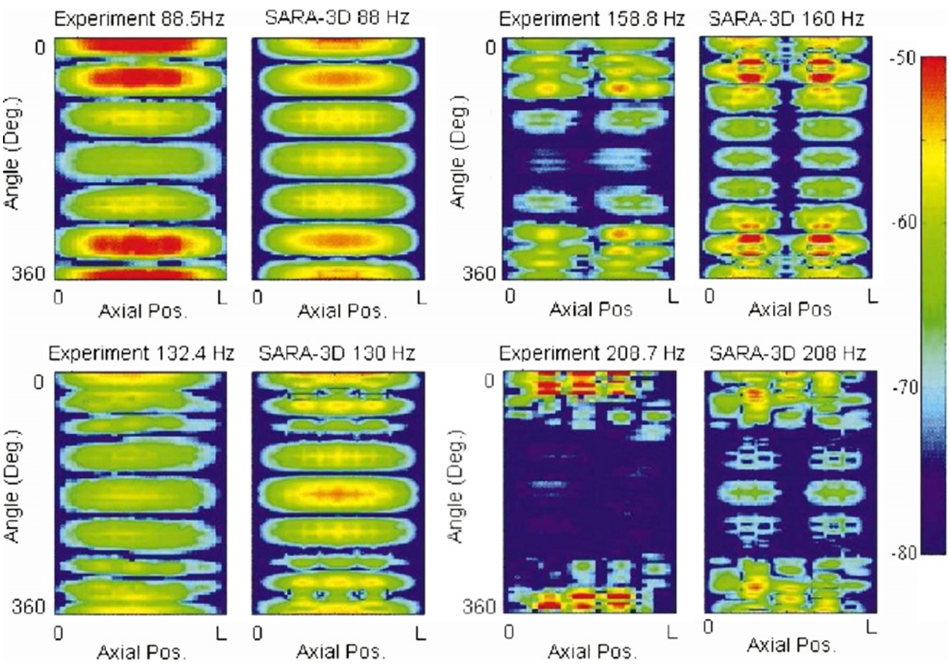

Fig. 5. Comparisons of the measured (left) and numerically predicted (right) magnitude in (m/s)/N of the normal surface velocity in circumferential angle vs. axial position for different frequencies. After [4].

that phase is nearly piecewise constant. But at the intermediate frequencies, the energy transfers become important from one substructure to another, as the vibrational energy is carried by their local modes. At higher frequencies, these transfers are smoothed out since the *diffusive regime* by which the energy amplitudes within the subsystems have stabilised, has been reached. Diffusion is manifested in the noisy pattern of the phase in this range. Note that the same conclusions were drawn in [3] from the analysis of the phase trends for the experimental structure of Fig. 1. Incidentally, these observations imply that smoothened amplitudes are not sufficient to describe mid-frequency vibrations, as one may be able to provide an information on the phase as well; this important point was also emphasised in [94].

2.2. Influence of uncertainties

The three frequency domains discussed in the previous section are characterised in the following way [3,95,96], provided that it is understood that the terms low-, mid- and high-frequency have a relative connotation requiring a more precise qualification on a case by case basis, for the particular system in consideration:

- A low-frequency response chiefly involves global, low-order structural eigenmodes which concentrate most of the vibrational energy.
- A mid-frequency response is characterised by the superposition of some global eigenmodes (as in the low frequency range) and clusters of local eigenmodes [97], which have an influence on both the local and global behaviours of the structure in the narrow frequency band where they are packed. Energy transfers between subsystems are important, and the phase can change significantly from one point to another in the overall system [98,99].
- At last, a high-frequency response involves numerous local eigenmodes which contribute to localise the vibrational energy. It is better characterised by quadratic observables smoothing out the contributions from the various local eigenmodes in given frequency bands, because the phase does not bring any additional information. Energy transfers between substructures are weak, may be negligible. The significance of choosing energy-type quantities in this range shall be addressed in more details in Section 3.3.2 below.

Because of the involvement of local modes, the predictive and irreducible uncertainties (as defined in [2]) and the structural complexity are two factors which influence heavily the structural responses in the mid- to high-frequency ranges. This may be shown invoking simple analytical arguments as well [3,100] (see also [101, Appendix 10] and [102]). Those factors chiefly arise for built-up structures having a repetitive pattern of any kind, but not necessarily periodic, as for example the above experimental structures or three-dimensional beam trusses [103]. Such structures actually have neighbouring, indeed merged multiple eigenfrequencies because they are constituted by assemblies of nearly identical subsystems exhibiting themselves comparable eigenfrequencies when they are considered in isolation. Multiple eigenfrequencies are attached to clusters of local eigenmodes for the different subsystems constituting the entire structure, which have a non-negligible, nay, dominant contribution to the overall vibrational energy in the higher frequency ranges.

More generally, the theory of symmetric, positive random matrices proposed by Soize [104–106] shows that the dispersion of structural dynamic responses increases when the frequency increases, for a fixed level of dispersion of the structural mass, stiffness and/or damping matrices. This theoretical result is confirmed by several experiments on engineering structures, notably in the automotive industry [107,108]. Fig. 6 for example displays the magnitudes of structure-borne frequency response functions measured by a microphone at the driver's head position for 99 samples of pickup trucks [108]. This example illustrates the main difficulties arising in the analyses of HF structural dynamics: the vibrational energy is spread over a large number of local, higher-order eigenmodes, none of them prevailing over the others in the overall system response, and the damping loss factors are small. It indicates that nominal responses obtained from a particular model are not particularly useful unless they are understood in a statistical sense, thus outlining the limitations of this model—especially in the HF range. These observations have prompted the development by engineers of the statistical energy analysis (SEA) of structural-acoustics systems. This global approach is briefly exposed in subsequent Section 2.3, before a more recent, though still heuristic local approach is introduced in Section 2.4.

2.3. Global approach: statistical energy analysis (SEA)

SEA characterises the HF dynamic properties of linear structures including the effects of uncertainties in energetic and statistical terms, hence the terminology. Indeed, the average mechanical energy within each substructure of the overall system constitutes the primal variable in the SEA formulation. This energy represents a response level integrated over a substructure, rather than a local estimation. The selection of subsystems should be guided by geometrical and/or physical (mechanical) considerations, provided that the assumptions upon which SEA is based are satisfied. For example, an assembly of two plates may be split into two substructures, one for each plate, but each of them may support several wave types – bending, shear, etc. – and therefore more than one SEA subsystem. The method is statistical in the sense that the mechanical parameters describing the dynamic response of the subsystems are uncertain and modeled by random

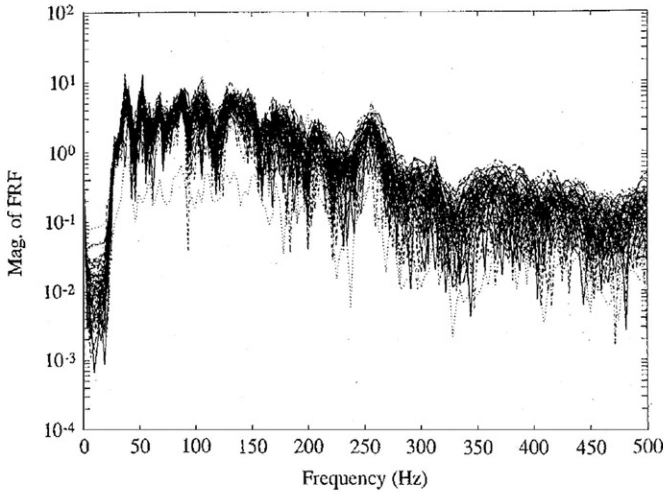

Fig. 6. Magnitude of 99 structure-borne frequency response functions of Isuzu pickup trucks: driver's head pressure for a mechanical impact excitation applied on the left front wheel. These measurements are scattered by a factor of 3 in the low-frequency range to a factor of about 10 for the higher frequencies. After [108].

variables. In its formal developments, however, SEA does not directly consider these parameters as being random, but rather uses some results of average equivalences for the vibrational energy computed from:

(i) the time average of the forced response of the substructures to some sustained deterministic loads;
(ii) or the mathematical expectation of this response to stationary random loads;
(iii) or the time average and mathematical expectation of this response to harmonic loads assuming that the eigenfrequencies are independent, uniform random variables.

Owing to these equivalences, the average energy within each substructure is re-interpreted as the ensemble average over a population of systems or industrial artefacts (typically the different samples of the same car type coming off a production line). Furthermore, it is evaluated over finite frequency bands which are broad enough to encompass a large number of eigenfrequencies, but narrow enough to emphasise the variation of response with frequency. Both processes of ensemble averaging and frequency integration tend to significantly reduce the variances of the measured responses, as observed in the previous examples. But since the response of a particular specimen has no reason to be the ensemble average of the parent population, this notion of variance or standard deviation should also be considered in the definition of the SEA ensemble response with respect to a mean or a trend. Tentative approaches have been suggested in the past [11] and more recently [109] assuming GOE (Gaussian orthogonal ensemble, see for example [54,106] and references therein) statistics of the natural frequencies.

SEA fundamental equations govern the relations between energy exchange and energy storage within subsystems, relying on an assumption of *weak coupling* between them. It favors a dynamical behaviour of the overall system such that its eigenmodes of vibration are close to the local eigenmodes of the subsystems considered in the isolation. In other words, the subsystems must be chosen in order to present substantial discontinuities of wave impedance, thereby satisfying the weak coupling assumption. That being so, the energy distribution over the system can only be decreasing as one moves away from the location of power input, in agreement with experiments. Thus SEA is a method dedicated to high frequencies but it cannot be extended to low frequencies as defined above. SEA basic equations are introduced in the following adopting a modal approach, in agreement with the model problem originally proposed in [6]. However, a wave approach may also be considered, see *e.g.* [2,58]. In collaboration with G. Maidanik [6] and then T.D. Scharton [110], R.H. Lyon has shown that the average power flow or energy flux (see [2]: both terms will be used indistinctly in this paper) $\mathbb{E}\{\Pi_{\alpha\beta}(t)\}$ between two linearly coupled, weakly damped oscillators identified by the indices α and β equal to either 1 or 2, is

$$\mathbb{E}\{\Pi_{\alpha\beta}(t)\} = \omega_\alpha\eta_{\alpha\beta}\mathbb{E}\{\mathcal{E}_{m\alpha}(t)\} - \omega_\beta\eta_{\beta\alpha}\mathbb{E}\{\mathcal{E}_{m\beta}(t)\}, \quad \alpha \neq \beta \in \{1,2\}, \tag{1}$$

provided they are subjected to uncorrelated broadband forces. That is to say, it is proportional to the difference of their average mechanical energies $\mathbb{E}\{\mathcal{E}_{m\alpha}(t)\}$. Here ω_α is the circular eigenfrequency of the αth oscillator considered in isolation, *i.e.* when the other oscillator $\beta \neq \alpha$ is "blocked". $\eta_{\alpha\beta}$ is the so-called coupling loss factor between both oscillators, which can be computed exactly as a function of the coupling parameters, see [11, Eq. (3.1.15)]. It satisfies the reciprocity (or consistency) relation $\omega_\alpha\eta_{\alpha\beta} = \omega_\beta\eta_{\beta\alpha}$ provided that the coupling is *conservative*—a core assumption in SEA modal approach. At last $\mathbb{E}\{\cdot\}$ stands for the time average, or the mathematical expectation, or the combination of both, but we shall not specify it in the following discussion by virtue of the principle of average equivalences invoked above. The power dissipated by each oscillator being $\Pi_{\text{dis},\alpha}(t) = \omega_\alpha\eta_\alpha\mathcal{E}_{m\alpha}(t)$ where η_α is the dissipation loss factor of the αth oscillator (twice its critical damping rate), the average energetic balance for each oscillator thus reads

$$\mathbb{E}\{\Pi_{\mathrm{in},\alpha}(t)\} = \omega_\alpha\eta_\alpha\mathbb{E}\{\mathcal{E}_{m\alpha}(t)\} + \omega_\alpha\eta_{\alpha\beta}(\mathbb{E}\{\mathcal{E}_{m\alpha}(t)\} - \mathbb{E}\{\mathcal{E}_{m\beta}(t)\}),$$

$\alpha \neq \beta \in \{1,2\}$, which constitutes the fundamental SEA equation for the coupling between the αth and $\beta \neq \alpha$ oscillators. It relates their average mechanical energies, forming the basic unknowns of the SEA method, with three of the fundamental mechanical parameters of this approach:

(i) the (small) dissipation loss factors η_α;
(ii) the coupling loss factors $\eta_{\alpha\beta}$;
(iii) the power inputs $\Pi_{\mathrm{in},\alpha}(t)$ induced by the applied loads.

In SEA it is subsequently assumed that this result extends to the conservative coupling of two continuous subsystems r and $s \neq r \in \{1,2\}$ considered as groups of N_r local eigenmodes only slightly modified by the coupling in a given frequency range of excitation $I_0 = [\omega_0 - \Delta\omega/2, \omega_0 + \Delta\omega/2]$. Let $\omega_{r,\alpha}$ and $\eta_{r,\alpha}$ be the circular eigenfrequencies and dissipation loss factors of the rth substructure when it is taken in isolation and is uncoupled from the other substructure, enforcing some boundary conditions on the interface as in *e.g.* [111]. Then let $\mathcal{I}_r = \{\alpha; \omega_{r,\alpha} \in I_0\}$ so that $N_r = \mathrm{card}\,\mathcal{I}_r$. The power exchanged between both substructures is written by analogy with Eq. (1)

$$\mathbb{E}\{\Pi_{rs}(t)\} \simeq \omega_0\eta_{rs}\mathbb{E}\{\mathcal{E}_{mr}(t)\} - \omega_0\eta_{sr}\mathbb{E}\{\mathcal{E}_{ms}(t)\}, \quad r \neq s \in \{1,2\}, \tag{2}$$

where $\mathcal{E}_{mr}(t) = M_r\sum_{\alpha\in\mathcal{I}_r}\dot{q}^2_{r,\alpha}(t)$, the sum of the contributions of the generalised coordinates $q_{r,\alpha}(t)$ for the response of the rth substructure projected on its local eigenmodes (normalised with respect to the mass; M_r being therefore the total mass of the rth subsystem) in the frequency range I_0. η_{rs}, η_{sr} are the average coupling loss factors of the substructures satisfying the reciprocity relation $N_r\eta_{rs} = N_s\eta_{sr}$. The latter involves the fourth fundamental mechanical parameter of SEA, the modal density $n_r(\omega_0) = N_r/\Delta\omega$. The average energetic balance for each subsystem is for $r \neq s \in \{1,2\}$

$$\mathbb{E}\{\Pi_{\mathrm{in},r}(t)\} = \omega_0\eta_r\mathbb{E}\{\mathcal{E}_{mr}(t)\} + \omega_0(\eta_{rs}\mathbb{E}\{\mathcal{E}_{mr}(t)\} - \eta_{sr}\mathbb{E}\{\mathcal{E}_{ms}(t)\}), \tag{3}$$

where $\eta_r(\omega_0)$ is the average dissipation loss factor of the rth substructure defined by $\omega_0\eta_r\mathcal{E}_{mr}(t) \simeq M_r\sum_{\alpha\in\mathcal{I}_r}\omega_{r,\alpha}\eta_{r,\alpha}\dot{q}^2_{r,\alpha}(t)$. The last step in the SEA formulation is to assume that Eq. (2) further applies to the weak, conservative coupling of $\mathcal{N}$ acoustical or mechanical subsystems. The average vibrational energies are then computed from the $\mathcal{N} \times \mathcal{N}$ matrix equation derived from Eq. (3) when it is written for each subsystem.

It may be seen from this discussion that SEA is based on successive assumptions, though it is observed in practice that this approach is rather effective. This is notably the case for the coupling of elastic structures with acoustic cavities, a situation where the hypotheses needed for these generalisations are actually fulfilled. Yet it should be noted that the approximation (2) above is in fact rigorously wrong as soon as the number of dofs exceeds two [112]. It is however rather well satisfied by reverberant subsystems, and this is the reason why SEA is a method dedicated to the HF range. Moreover, it does not have any straightforward extension to the transient regime.

2.4. Local approach: continuity equation and vibrational conductivity analogy (VCA)

Besides this limitation, SEA does not give any information more precise than the average vibrational energy integrated over substructures. Approaches based on an analogy with heat conduction, referred to as vibrational conductivity analogy (VCA) [16–24], are aimed at describing locally how this energy is spread. They are derived from a standard continuity equation (local balance of energy) in elastodynamics including dissipation phenomena. This derivation is detailed below, starting from the equilibrium and constitutive equations pertaining to structural dynamics. Several assumptions will be introduced in the course of the analysis, which limit the range of applicability of VCA. The purpose of this section is thus to explain how VCA equations are obtained, and then underline these limitations in order to motivate the developments presented in Sections 3 and 4. Here a possible strategy for the improvement of SEA or VCA modeling is proposed, based on the restrictions identified for those existing engineering approaches. This section also introduces some basic notations used subsequently in Section 3.

Let $\mathcal{O} \subseteq \mathbb{R}^d$ be an open domain occupied by an heterogeneous linear viscoelastic material, where $d = 1,2$ or 3 is the physical space dimension. Its density is denoted by $\varrho(\mathbf{x})$ and its fourth-order relaxation tensor is denoted by $\mathbf{C}(\mathbf{x},t)$, $\mathbf{x} \in \mathcal{O}$, $t \in \mathbb{R}_+$. We consider the vibrations of that medium about a static equilibrium configuration considered as its natural state (neglecting prestress) when it is subjected to HF initial conditions and body forces. The latter are characterised by their highly oscillatory feature, which is embodied in a small parameter $\varepsilon > 0$. It is for example the rate of variation of these loads with respect to the typical size of the medium, but it could be any other parameter which recall their HF content; there is no need to clarify what it is exactly for the subsequent analysis. Then the displacement field $\mathbf{u}(\mathbf{x},t)$ of the medium about its reference configuration will be parameterised by ε as well, and will have the same oscillatory feature as the loads which have generated it. Thus let $\boldsymbol{\sigma}(\mathbf{x},t)$ be the second-order Cauchy stress tensor of the medium. The balance of momentum in a fixed reference frame considering the action of body forces given by their density $\varrho(\mathbf{x})\mathbf{f}(\mathbf{x},t;\varepsilon)$ is

$$\varrho\partial_t^2\mathbf{u} = \mathbf{Div}_{\mathbf{x}}\boldsymbol{\sigma} + \varrho\mathbf{f}, \quad \mathbf{x},t \in \mathcal{O} \times \mathbb{R}. \tag{4}$$

Here the divergence of a second-order tensor $\mathbf{A}$ is defined by $(\mathbf{Div_x A}, \mathbf{b}) = \nabla_\mathbf{x} \cdot (\mathbf{A}^\mathsf{T}\mathbf{b})$ for any constant vector $\mathbf{b}$, and $\nabla_\mathbf{x}$ or more simply ∇ if no ambiguity holds is the gradient vector with respect to the point $\mathbf{x} \in \mathbb{R}^d$. At last $(\cdot)^\mathsf{T}$ stands for the matrix transpose. Eq. (4) is supplemented by the initial conditions

$$\mathbf{u}(\mathbf{x},0) = \mathbf{u}_0(\mathbf{x};\varepsilon), \quad \partial_t \mathbf{u}(\mathbf{x},0) = \mathbf{v}_0(\mathbf{x};\varepsilon), \tag{5}$$

which are, as explained above, parameterised by ε as well as the body forces $\mathbf{f}$. For example, the classical WKB initial conditions (HF plane waves or geometric optics) $\mathbf{u}_0(\mathbf{x};\varepsilon) = \varepsilon \mathbf{A}(\mathbf{x}) \mathrm{e}^{\mathrm{i}\mathbf{k}\cdot\mathbf{x}/\varepsilon}$ and $\mathbf{v}_0(\mathbf{x};\varepsilon) = \mathbf{B}(\mathbf{x}) \mathrm{e}^{\mathrm{i}\mathbf{k}\cdot\mathbf{x}/\varepsilon}$, for a wave vector $\mathbf{k} \in \mathbb{R}^d$ and smooth compactly supported amplitudes $\mathbf{A}(\mathbf{x})$ and $\mathbf{B}(\mathbf{x})$, correspond to an initial motion oscillating at a spatial period proportional to $\varepsilon > 0$. Indeed such data have a rapidly oscillating phase $(\mathbf{k} \cdot \mathbf{x})/\varepsilon$ whenever $\varepsilon \ll 1$, and slowly varying amplitudes. From the standard properties of wave equations they will give rise to HF waves within the medium oscillating at the same rate ε for all times $t > 0$. Many other choices are of course possible (see for example [33]) provided that these data remain strongly oscillatory with respect to ε [26,28] that is, they oscillate on a scale which is not smaller than ε.

The stress field $\boldsymbol{\sigma}(\mathbf{x},t)$ on $\overline{\mathcal{O}} \times \mathbb{R}$ is given as a function of the linearised strain tensor $\boldsymbol{\epsilon}(\mathbf{x},t)$ by the material constitutive equation

$$\boldsymbol{\sigma}(\mathbf{x},t) = \int_{-\infty}^{t} \mathbf{C}(\mathbf{x},t-\tau)\partial_t \boldsymbol{\epsilon}(\mathbf{x},\tau)\,\mathrm{d}\tau, \quad \boldsymbol{\epsilon}(\mathbf{x},t) = \nabla_\mathbf{x} \otimes_s \mathbf{u}(\mathbf{x},t).$$

Here $\otimes_s$ is the symmetrised tensor product of two vectors $\mathbf{a} \otimes_s \mathbf{b} = \mathrm{sym}(\mathbf{a} \otimes \mathbf{b})$. If the motion starts at time $t=0$, and if $\boldsymbol{\epsilon} = \boldsymbol{\sigma} = \mathbf{0}$ for $t < 0$, the constitutive equation becomes

$$\begin{aligned}\boldsymbol{\sigma}(\mathbf{x},t) &= \mathbf{C}(\mathbf{x},t)\boldsymbol{\epsilon}(\mathbf{x},0) + \int_0^t \mathbf{C}(\mathbf{x},t-\tau)\partial_t \boldsymbol{\epsilon}(\mathbf{x},\tau)\,\mathrm{d}\tau = \mathbf{C}(\mathbf{x},0)\boldsymbol{\epsilon}(\mathbf{x},t) + \int_0^t \partial_t \mathbf{C}(\mathbf{x},\tau)\boldsymbol{\epsilon}(\mathbf{x},t-\tau)\,\mathrm{d}\tau \\ &= \partial_t\left(\int_0^t \mathbf{C}(\mathbf{x},\tau)\boldsymbol{\epsilon}(\mathbf{x},t-\tau)\,\mathrm{d}\tau\right), \quad \forall t \geq 0, \quad \forall \mathbf{x} \in \overline{\mathcal{O}}.\end{aligned} \tag{6}$$

It is usually not possible to define in a non-ambiguous way the free (stored) and dissipated energies with such a constitutive relation, unless it can be expressed in terms of internal variables and the relaxation function has an exponential form. Zener constitutive model for example

$$\boldsymbol{\sigma} + \tau_0 \partial_t \boldsymbol{\sigma} = \mathbf{C}^e \boldsymbol{\epsilon} + \tau_0 \mathbf{C}^\nu \partial_t \boldsymbol{\epsilon},$$

where $\tau_0 > 0$ is a relaxation time, and $\mathbf{C}^e$ and $\mathbf{C}^\nu$ are fourth-order symmetric tensors, corresponds to

$$\mathbf{C}(\mathbf{x},t) = \mathbf{C}^e(\mathbf{x}) + \mathrm{e}^{-t/\tau_0}(\mathbf{C}^\nu(\mathbf{x}) - \mathbf{C}^e(\mathbf{x})),$$

provided that $\mathbf{C}^\nu - \mathbf{C}^e$ is positive definite in order to actually obtain a dissipative model. Then the following definitions of the sum of the free and kinetic energy densities $\mathcal{E}(\mathbf{x},t)$, a positive scalar, and of the energy flux density $\boldsymbol{\pi}(\mathbf{x},t)$, or the Poynting vector of the viscoelastic medium, can be chosen

$$\mathcal{E} = \tfrac{1}{2}(\varrho|\partial_t \mathbf{u}|^2 + \mathbf{C}^e \boldsymbol{\epsilon} : \boldsymbol{\epsilon} + (\mathbf{C}^\nu - \mathbf{C}^e)^{-1}\boldsymbol{\tau} : \boldsymbol{\tau}), \quad \boldsymbol{\pi} = -\boldsymbol{\sigma}\partial_t \mathbf{u}, \tag{7}$$

where $\boldsymbol{\tau} := \boldsymbol{\sigma} - \mathbf{C}^e \boldsymbol{\epsilon}$, and $\mathbf{A} : \mathbf{B} = \mathrm{Tr}(\mathbf{A}\mathbf{B}^\mathsf{T})$ stands for the scalar product of the second-order tensors $\mathbf{A}$ and $\mathbf{B}$. Here $\mathrm{Tr}\,\mathbf{A}$ is the trace of a square matrix $\mathbf{A}$. As for the dissipated and power input densities $\pi_{\mathrm{dis}}(\mathbf{x},t)$ and $\pi_{\mathrm{in}}(\mathbf{x},t)$, respectively, they are defined by

$$\pi_{\mathrm{dis}} = \frac{1}{\tau_0}(\mathbf{C}^\nu - \mathbf{C}^e)^{-1}\boldsymbol{\tau} : \boldsymbol{\tau}, \quad \pi_{\mathrm{in}} = \varrho \mathbf{f} \cdot \partial_t \mathbf{u}. \tag{8}$$

These quantities satisfy the continuity equation

$$\partial_t \mathcal{E} + \mathrm{div}_\mathbf{x} \boldsymbol{\pi} + \pi_{\mathrm{dis}} = \pi_{\mathrm{in}}. \tag{9}$$

It is obtained by multiplying Eq. (4) by $\partial_t \mathbf{u}$ and then using the definitions (7) and (8) for Zener's constitutive equation. If the medium is purely elastic (memoryless material), *i.e.* $\mathbf{C}$ does not depend on time ($\partial_t \mathbf{C} = \mathbf{0}$, corresponding to $\tau_0 = 0$ in Zener viscoelastic model), the third term in the above equation vanishes. If in addition the power input is zero, a usual conservation equation is derived as $\partial_t \mathcal{E} + \mathrm{div}_\mathbf{x} \boldsymbol{\pi} = 0$; besides

$$\int_{\mathcal{O}} \mathcal{E}(\mathbf{x},t)\,\mathrm{d}\mathbf{x} = \int_{\mathcal{O}} \mathcal{E}(\mathbf{x},0)\,\mathrm{d}\mathbf{x}, \quad \forall t \geq 0.$$

Eq. (9) is the local counterpart of Eq. (3) of SEA for the balance of power between subsystems. Indeed, if (9) is integrated over a bounded subdomain $\mathcal{D}_r$ of $\mathcal{O}$, then invoking Ostrogradski's formula yields

$$\int_{\mathcal{D}_r} \pi_{\mathrm{in}}\,\mathrm{d}\mathbf{x} = \partial_t \int_{\mathcal{D}_r} \mathcal{E}\,\mathrm{d}\mathbf{x} + \int_{\mathcal{D}_r} \pi_{\mathrm{dis}}\,\mathrm{d}\mathbf{x} + \int_{\partial\mathcal{D}_r} \boldsymbol{\pi} \cdot \hat{\mathbf{n}}_r \mathrm{d}\gamma,$$

$\hat{\mathbf{n}}_r$ being the outward unit normal to $\mathcal{D}_r$. In the harmonic regime $\mathbf{u}(\mathbf{x},t) = \widehat{\mathbf{u}}(\mathbf{x},\omega_0)\mathrm{e}^{\mathrm{i}\omega_0 t}$ such that $\partial_t \mathcal{E} = 0$. The third term at the right-hand side above is exactly the power exchanged by $\mathcal{D}_r$ with its neighbourhood, given as the difference of their mechanical energies in SEA. VCA approaches as introduced by Nefske and Sung [17] have a direct link with both formulations. Indeed, these authors have proposed to write the energy flow

$$\widehat{\boldsymbol{\pi}} = -D(\omega_0)\nabla_\mathbf{x} \widehat{\mathcal{E}} \tag{10}$$

invoking a Fourier-like, or Fick-like law as the frequency increases. $D(\omega_0)$ is a diffusion coefficient depending on the (large) central circular frequency ω_0 of the frequency range I_0 of the loads in a steady-state regime. In this respect, (10) is a local counterpart of (2), whereas (9) becomes a diffusion equation yielding, once it is integrated over the subdomain $\mathcal{D}_r$

$$\int_{\mathcal{D}_r} \widehat{\pi}_{\text{in}}\, \mathrm{d}\mathbf{x} = \int_{\mathcal{D}_r} \widehat{\pi}_{\text{dis}}\, \mathrm{d}\mathbf{x} - \int_{\partial\mathcal{D}_r} D(\omega_0)\partial_{\hat{\mathbf{n}}_r}\widehat{\mathcal{E}}\, \mathrm{d}\gamma$$

to be compared with (3). VCA approaches have only been applied to simple systems until now, namely *homogeneous* beams and thin plates, since they are based, as seen here, on the assumption (10). Although it may have a rather firm theoretical basis for one-dimensional waveguides (rods), this conjecture is very restrictive, and even gets wrong for other structures or higher physical dimensions ($d=2$ or 3, see for instance the discussions in [20,21]). This issue will be addressed further on in Section 3.4.

2.5. Research needs and objectives

The previous presentation of SEA and VCA has shown that both approaches rely on some crucial assumptions, namely (2) and (10) respectively, though they may be very effective in the current engineering practice. This observation motivates the consideration of an alternative point of view, which should be able to release the limitations imposed in SEA and VCA derivations. It is argued in the remaining of the paper that *kinetic models* of wave propagation phenomena may be well adapted to this objective. This is because they describe highly oscillating waves adopting an energetic point of view in the phase space position $\times$ wave vectors, and they apply to heterogeneous (possibly random) media. They also allow to derive rigorously diffusion limits for multiply scattered waves, a situation pertaining to the engineering appraisal of HF vibrations. These models are described in Section 3 in the context of the different regimes applicable to wave propagation phenomena, depending of the scaling of the wavelength vs. a typical size of the inhomogeneities. The different results presented there apply to open, non-dissipative media. The implementation of kinetic models in structural dynamics thus requires two important extensions: (i) the consideration of dissipative phenomena, paying particular attention to the construction of models consistent with the use of energetic observables; and (ii) the consideration of dedicated boundary conditions for such quantities, consistent with the boundary conditions imposed to the underlying wave fields. Both issues are addressed in Section 4. Damping modeling and boundary conditions for kinetic equations are outlined in Sections 4.1 and 4.2, respectively, and an overall assessment of the various developments presented in the paper is given in Section 4.3. The objectives of this research are twofold. As regards modeling issues on one hand, kinetic models are aimed at justifying and possibly extending the SEA and VCA approaches on the basis of a firm, more general theoretical ground and weakened assumptions. It is also focused on the transient domain for which the existing literature and results are rather sparse. Regarding the applications on the other hand, the proposed models are dedicated to the computation of the dynamic responses of built-up structures impacted by mechanical, acoustical or aerodynamic loads either in the transient or the steady-state regimes. Examples include cars, railway trains, aircrafts, spacecrafts, ships, pipelines, buildings, or industrial plants.

3. High-frequency elastic wave propagation in heterogeneous media

Electromagnetic, acoustic or elastic waves in an homogeneous medium have a fairly well identified behaviour, even in the presence of a smooth boundary as a free surface for example. This is no longer the case in an heterogeneous medium or a piecewise homogeneous medium, not to mention a random medium. Wave propagation in heterogeneous media is the subject of intense, multidisciplinary researches in order to exhibit, on one hand, some new phenomena such as anisotropic diffusion (strong localisation) or coherent back-scattering effects (weak localisation), and to take advantage, on the other hand, of these effects in applications such as time reversal techniques. This issue is less studied in continuum mechanics, beside some early works in seismology and surface geophysics [113]. However, it is essential in the understanding of the HF vibrational phenomena described in the previous section. The purpose of this section is therefore to present the different approaches invoked in modeling wave propagation in heterogeneous media, although such a description cannot pretend to be exhaustive in view of the considerable existing literature on the subject. Refs. [46–48] for example, so as to cite only recent works, develop a thorough and much more relevant analysis than the simple ideas expounded below. These different approaches are emphasised here in view of introducing the kinetic modeling strategy adopted for the description of HF vibrations of slender structures, which is explained in subsequent Section 4.

3.1. Characteristic scales

The identification of wave propagation phenomena in heterogeneous media is based on the formal identification of several characteristic scales: the wavelength λ, the characteristic length of heterogeneities ℓ in the medium or their correlation length ℓ_c if they are random, and the size of the sample, or the observation/propagation distance L. These scales allow to gradually distinguish local to global effects. One can then identify the so-called microscopic, mesoscopic, and macroscopic regimes depending on, roughly speaking, the ratio of the wavelength λ to the scale ℓ, $\delta=\lambda/\ell$; one also introduces $\kappa=\ell/L$ or ℓ_c/L (the Knudsen number) in the subsequent analyses. These different regimes are characterised as follows:

- *Microscopic regime* $\delta>1$: The relevant evolution model is the wave equation. The location and shape of all heterogeneities, inclusions, holes, boundaries, interfaces, stiffeners in a structure, have to be known precisely. If distances between these heterogeneities are comparable to the wavelength, resonance phenomena basically arise.
- *Mesoscopic regime* $\delta\sim 1$: The relevant evolution model is the transport equation, or the radiative transfer (linear Boltzmann) equation in a "high-frequency" random medium where the correlation length is comparable to the wavelength. The characteristic parameters of this medium are its scattering mean free paths or the transport velocities, which are derived from the microscopic regime. At these scales complex weak or strong localisation phenomena may arise, as a result of the interaction of scattered waves with heterogeneities. These phenomena will not be addressed in this discussion.
- *Macroscopic regime* $\delta<1$: The relevant evolution model is the diffusion equation. It is characterised by a diffusion coefficient which is derived from the mesoscopic parameters. In this regime all interferences of waves with heterogeneities are smoothed out and angularly averaged intensities are considered.

A fourth scale, the absorption length L_a for optical systems or the incoherence length L_{i} for electronic systems, is also introduced by physicists. The whole Section 3 is focused on non-dissipative media, and weakly dissipative structures are considered in Section 4.1; thus one always has $L_a=+\infty$ or at least $L\ll L_a$, and this scale is ignored.

Of course several characteristic lengths and wavelengths can coexist. The following discussion is only a crude and naive simplification of a much more complex reality where different phenomena interact at different scales, described by different, possibly competing models. Another difficulty is raised by the polarised feature of elastic waves: if materials are isotropic, they are constituted by a single compressional mode (denoted by P) similarly to acoustic waves, and a twofold shear mode (denoted by S) similarly to electromagnetic waves, with possible conversions between both modes. This additional complexity has a direct influence on the understanding of the various phenomena to be identified, because they may arise for P and S waves simultaneously. Finally, it is also of note to mention that the behaviour of the waves near and at the transition between the different regimes outlined above is not well understood, and the models should ideally be improved in order to explain it. Now the microscopic, mesoscopic and macroscopic regimes are reviewed in the next three sections.

3.2. Microscopic regime

If ℓ is the scale of variations of the mechanical parameters of the medium and if these variations can be described fairly well, then as soon as λ is greater than ℓ the propagation regime is the low-frequency microscopic regime; the relevant evolution model is the balance equation of elastodynamics in $\mathcal{O}\times\mathbb{R}$

$$\varrho\partial_t^2\mathbf{u}_\varepsilon=\boldsymbol{\nabla}_{\mathbf{x}}\cdot(\mathbf{C}^e:\boldsymbol{\nabla}_{\mathbf{x}}\otimes\mathbf{u}_\varepsilon)+\varrho\mathbf{f}_\varepsilon. \tag{11}$$

Eq. (11) is nothing but Eq. (4) ignoring dissipation. Also the displacement field $\mathbf{u}$ has been indexed by ε in order to recall that it is parameterised by this parameter since the applied loads (initial conditions (5) and body forces $\mathbf{f}$) are; typically $\varepsilon=\lambda/L$, the normalised wavelength of the waves they generate within the medium. The dominant physical phenomena in this case are (i) single scattering, corresponding to the interaction of an incident wave with a bounded, well identified inhomogeneity backscattering waves at different frequencies, directions and polarisations, and (ii) resonance, as a result of constructive interferences at particular frequencies. In soil–structure interaction for example resonant modes almost identical to the structural modes *in vacuo* arise. However, some specific interaction modes may appear as well, for the case of stiff media coupled to a flexible one, or modes related to the system shape (a stratified medium, typically, for a composite structure or a soil); see [114] and references therein.

The limit case $\delta\gg 1$, where mechanical parameters of the medium vary rapidly, can be treated by homogenisation techniques or an effective medium theory; see for example [115,116] for applications in elastodynamics. Both approaches apply to the case of randomly distributed inclusions of which number increases indefinitely at a constant volumic fraction in an homogeneous background medium. They are relevant as long as the propagation distance L remains comparable to the wavelength, $\varepsilon\sim 1$. They are very effective for steady-state problems in enclosed area, but they do not account for propagation and transport effects (of the associated energy) when $\varepsilon\ll 1$. This is the mesoscopic regime outlined in the next section.

3.3. Mesoscopic regime

For short wavelengths with respect to the observation distance within the propagation medium, considering first that its mechanical parameters vary slowly, the relevant evolution model is the transport equation. It describes the mesoscopic regime of propagation, corresponding to the mid- to high-frequency vibrations of bounded structures. Therefore, the subsequent presentation introduces the main notations and concepts which will be used in Section 4 to elaborate a transient transport model for the response of slender structures to high-frequency excitations. It starts with a short survey of classical WKBJ, or ray methods which is primarily aimed at expounding the notations and the Lagrangian approach of wave front tracking for short wavelengths. The latter is formally generalised in Sections 3.3.2 and 3.3.3 which summarise the main concepts invoked in kinetic models.

3.3.1. Ray methods

The transport model is known in room acoustics [117] or seismology [71,72], among others, in the form of the ray theory:

(i) with a real phase, or WKBJ method, after the physicists G. Wentzel, H. Kramers and L. Brillouin, and the mathematician H. Jeffreys who independently formalised it in 1920s; a first evidence can be found in the early works of F. Carlini (1817) or G. Green and J. Liouville (1837) as reviewed in [70];
(ii) or with a complex phase, for example the method of Gaussian beams [118], widely used in elastodynamics [119], underwater acoustics [120], or electromagnetism [121].

It consists in seeking a solution of Eq. (11) in the form

$$\mathbf{u}_\varepsilon(\mathbf{x},t) \simeq \mathrm{e}^{\mathrm{i}S(\mathbf{x},t)/\varepsilon}\sum_{k=0}^{\infty}\varepsilon^k\mathbf{U}_k(\mathbf{x},t), \quad \varepsilon = \frac{\lambda}{L} \ll 1, \tag{12}$$

where the phase function $S(\mathbf{x},t)$ is either real in the WKBJ method, or complex for a Gaussian beam. Plugging the ansatz (12) into Eq. (11), it is shown, adopting an Eulerian point of view (see Fig. 7(b)), that the phase function satisfies an eikonal equation and that the density $|\mathbf{U}_0|^2$ satisfies a linear transport equation of which coefficients depend on the phase function. Indeed, starting from (11) in the homogeneous case $\mathbf{f}_\varepsilon \equiv \mathbf{0}$ (the elastic wave equation), the acoustic, or Christoffel tensor $\boldsymbol{\Gamma}(\mathbf{x},\mathbf{k})$, and the dispersion matrix $\mathbf{H}(\mathbf{s},\xi)$ of the propagation medium are introduced as follows:

$$\boldsymbol{\Gamma}(\mathbf{x},\mathbf{k})\mathbf{U} = \varrho(\mathbf{x})^{-1}(\mathbf{C}^e(\mathbf{x}):\mathbf{U}\otimes\mathbf{k})\mathbf{k}, \quad \mathbf{U}\in\mathbb{R}^n, \tag{13a}$$

$$\mathbf{H}(\mathbf{s},\xi) = \varrho(\mathbf{x})(\boldsymbol{\Gamma}(\mathbf{x},\mathbf{k}) - \omega^2\mathbf{I}_n), \tag{13b}$$

where $\mathbf{s} = (\mathbf{x},t)$, $\xi = (\mathbf{k},\omega)$, $(\mathbf{s},\xi) \in T^*(\mathcal{O}\times\mathbb{R}) := \mathcal{O}\times\mathbb{R}_t\times\mathbb{R}^d_{\mathbf{k}}\times\mathbb{R}_\omega$, and $\mathbf{I}_n$ is the identity matrix of $\mathbb{R}^n$. Here n is the dimension of the wave system, which is d in elastodynamics but may be greater than the physical space dimension for wave systems arising from reduced kinematics as with thick beams or shells. $\mathbf{H}$ does not depend on the scale ε, that is to say we consider high-frequency wave propagation $\varepsilon \ll 1$ in a slowly varying "low frequency" medium $\kappa \sim 1$. The eikonal and zeroth-order transport read

$$\mathbf{H}(\mathbf{s}\boldsymbol{\nabla}_\mathbf{s}S)\mathbf{U}_0 = \mathbf{0}, \tag{14a}$$

$$\boldsymbol{\nabla}_\mathbf{s}\cdot(\mathbf{U}_0^{\mathsf{T}}\boldsymbol{\nabla}_\xi\mathbf{H}(\mathbf{s},\boldsymbol{\nabla}_\mathbf{s}S)\mathbf{U}_0) = 0, \tag{14b}$$

respectively. Adopting a Lagrangian point of view (see Fig. 7(c)), the pair $(\mathbf{s},\boldsymbol{\nabla}_\mathbf{s}S)$ is the solution of the Hamiltonian system associated to the elastic wave equation, which consists in solving the eikonal equation by the method of characteristics, or ray tracing. Thus introducing the Hamiltonian $\mathcal{H} = \det\mathbf{H}$, the usual properties of the acoustic tensor $\boldsymbol{\Gamma}$ are such that

$$\mathcal{H}(\mathbf{s},\xi) = \prod_{\alpha=1}^{n}\mathcal{H}_\alpha(\mathbf{s},\xi), \quad \mathcal{H}_\alpha(\mathbf{s},\xi) = \varrho(\mathbf{x})(\lambda_\alpha^2(\mathbf{x},\mathbf{k}) - \omega^2), \tag{15}$$

where λ_α^2 stands for the αth (positive) eigenvalue of $\boldsymbol{\Gamma}$ with $1\le\alpha\le n$. Considering isotropic elasticity for instance, $\boldsymbol{\Gamma}(\mathbf{x},\mathbf{k}) = \lambda_P^2(\mathbf{x},\mathbf{k})\hat{\mathbf{k}}\otimes\hat{\mathbf{k}} + \lambda_S^2(\mathbf{x},\mathbf{k})(\mathbf{I}_d - \hat{\mathbf{k}}\otimes\hat{\mathbf{k}})$ and $\lambda_\alpha(\mathbf{x},\mathbf{k}) = c_\alpha(\mathbf{x})|\mathbf{k}|$ of multiplicity 1 if $\alpha = P$ or 2 if $\alpha = S$, c_P and c_S being the elastic compressional and shear wave velocities, respectively, such that $c_S < c_P$. It is assumed subsequently that the eigenvalues of the acoustic tensor are distinct. The case of multiple eigenvalues with constant multiplicity, as for isotropic elasticity, is dealt with along the same lines although the writing is more involved; see [26,28,34]. So for systems such that $n \ge d$, n eigenvalues shall be considered altogether, counting for their orders of multiplicity.

Then the corresponding Hamiltonian equations for $1\le\alpha\le n$ are

$$\begin{aligned}\frac{\mathrm{d}\mathbf{s}}{\mathrm{d}\tau} &= \boldsymbol{\nabla}_\xi\mathcal{H}_\alpha(\mathbf{s}(\tau),\xi(\tau)), \quad \mathbf{s}(0) = \mathbf{s}_0,\\ \frac{\mathrm{d}\xi}{\mathrm{d}\tau} &= -\boldsymbol{\nabla}_\mathbf{s}\mathcal{H}_\alpha(\mathbf{s}(\tau),\xi(\tau)), \quad \xi(0) = \xi_0 \ne \mathbf{0},\end{aligned} \tag{16}$$

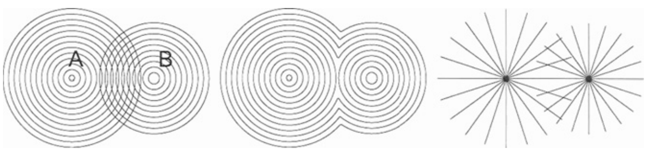

Fig. 7. Ray method with real phase for acoustic wave propagation with two monopoles A and B: exact solution (a), viscosity solution (b), and ray tracing (c). After [73].

in $T^*(\mathcal{O}\times\mathbb{R})\backslash\{(\mathbf{s},\xi);\ \xi=\mathbf{0}\}$, with an initial condition $(\mathbf{s}_0,\xi_0)$ satisfying $\mathcal{H}_\alpha(\mathbf{s}_0,\xi_0)=0$. Note that it follows from this condition that $\mathcal{H}_\alpha(\mathbf{s}(\tau),\xi(\tau))=0$ for all τ. The *rays* strictly speaking are defined as the projections on $\mathcal{O}\times\mathbb{R}_t$ of the bicharacteristic curves $\tau\mapsto(\mathbf{s}_\alpha(\tau),\xi_\alpha(\tau))$ solving (16), that is $\tau\mapsto\mathbf{s}_\alpha(\tau)$. These rays are space–time curves parameterised by τ (a curvilinear abscissa, or a time in $\mathbb{R}$) associated to propagation *modes*, or polarisations α defined by the dispersion relations $\mathcal{H}_\alpha(\mathbf{s},\xi)=0$. This interpretation extends to randomly perturbed homogeneous media considering (16) as a system of stochastic differential equations for a transport process $\{(\mathbf{S}_\tau,\Xi_\tau),\tau\geq 0\}$ [122]. The ray method with a real phase has some major drawbacks from either an Eulerian point of view or a Lagrangian point of view. The nonlinearity of the eikonal equation does not allow to superpose different phases, as required in Fig. 7(a). One way to circumvent this difficulty is to consider the notion of viscosity solution [123], as in Fig. 7(b). From the Lagrangian point of view, ray tracing is no longer possible on the caustics, where the rays stack, because the amplitudes $\mathbf{U}_k$ rapidly increase in their neighbourhood, and even blow up on the rays themselves. Using a complex phase function $S(\mathbf{x},t)$ with real values on the rays solely, and such that the imaginary part of its Hessian $\nabla_\mathbf{x}\otimes\nabla_\mathbf{x}S$ is positive definite on the rays [118], is a mean to release these restrictions. Indeed, if $(\mathbf{s}(\tau),\mathbf{k}(\tau))$ stands for a solution of the associated Hamiltonian system, and if $\mathbf{y}$ is orthogonal to $\mathbf{x}(\tau)$ and $\nabla_\mathbf{x}S(\mathbf{s}(\tau))=\mathbf{k}(\tau)$, taking the Taylor expansion of the phase about a ray for $|\mathbf{y}|\ll|\mathbf{x}(\tau)|$ yields

$$\mathbf{u}_\varepsilon(\mathbf{x}(\tau)+\mathbf{y},t(\tau))\approx\mathbf{U}_0(\mathbf{s}(\tau))\mathrm{e}^{\mathrm{i}\{S(\mathbf{s}(\tau))+(1/2)\mathbf{y}^\mathsf{T}[\nabla_\mathbf{x}\otimes\nabla_\mathbf{x}S(\mathbf{s}(\tau))]\mathbf{y}\}/\varepsilon} \tag{17}$$

as an approximate solution of (11). It is concentrated on the ray, but also defined away from it with a Gaussian profile which narrows when the frequency increases (or $\varepsilon\to 0$). The advantage of this ansatz is that it is no longer necessary to satisfy the eikonal equation exactly, but only its Taylor expansion up to a given order controlling the accuracy of the approximate solution. It constitutes an alternative approach to the usual WKBJ method in order to describe the rays away from the caustics and globally in time.

Applications of such ray methods for elastic wave propagation in open media or slender structures are too numerous to be listed here exhaustively; however, some classical and more recent references are (but not limited to) [69–81,119–121,124].

3.3.2. Transport model

The ansatz (12) of the ray theory is only one *a priori* particular construction of high-frequency solutions of the elastic wave equation. This approach also requires rather strong regularity assumptions for the initial conditions of the phase S and amplitude $\mathbf{U}_0$. The more recent works of Tartar [125], Gérard et al. [26,126], Lions and Paul [25], Papanicolaou and Ryzhik [28], or Bal [30] on the microlocal analysis of wave systems have generalised this theory for weaker assumptions on their high-frequency solutions and the initial conditions. These authors have shown that the energy density associated to *all* oscillating solutions (not only those having the form of Eq. (12)), resolved in the phase space position $\times$ wave vector, satisfies a Liouville-type transport equation. These results are now well established in physics [39–48], but have been less considered in the engineering mechanics or engineering materials community [49–54]. The main mathematical tool for the derivation of a transport equation from a wave equation is the Wigner transform, of which high-frequency limit $\varepsilon\to 0$, the so-called Wigner measure, captures the vibrational energy density in phase space. The advantage of this new representation is that it clears all classical difficulties inherited from ray methods, and it yields global propagation properties of the energy for weakened regularity assumptions of the initial conditions [127]. In return, the explicit knowledge of the phase is lost. The eikonal equation is replaced by the dependence of the Wigner measure vs. ξ, which gives its propagation directions as obtained from the dispersion equation $\mathcal{H}(\mathbf{s},\xi)=0$.

Going back to the system (16) and considering its solutions $\tau\mapsto(\mathbf{s}(\tau),\xi(\tau))=\Phi_\tau(\mathbf{s}_0,\xi_0)$ as the paths in phase space of some energy "particles" with an overall density denoted by $W(\mathbf{s}(\tau),\xi(\tau))$, then

$$\frac{\mathrm{d}W}{\mathrm{d}\tau}=\{\mathcal{H},W\}=0, \tag{18}$$

where $\{f,g\}:=\nabla_\xi f\cdot\nabla_\mathbf{s}g-\nabla_\mathbf{s}f\cdot\nabla_\xi g$ is the usual Poisson bracket. Eq. (18) is the Liouville equation which is the expression of the conservation of W in phase space starting from the initial data $W(\mathbf{s}_0,\xi_0):=W_0(\mathbf{s},\xi)$. As the dispersion matrix $\mathbf{H}$ is independent of time, Eq. (16) yields $\mathrm{d}\omega/\mathrm{d}\tau=0$ and W has on $\mathcal{X}:=T^*(\mathcal{O}\times\mathbb{R})\backslash\{\mathbf{k}=\mathbf{0}\}$ the form

$$W=\sum_{\alpha=1}^{n}W_\alpha\delta(\mathcal{H}_\alpha), \tag{19}$$

where $\delta(\cdot)$ is the usual Dirac measure, and the W_α's are referred to as *specific intensities* in the dedicated literature. They characterise the amount of energy density concentrated on the different rays existing for the different polarisation modes α given by the dispersion relations $\mathcal{H}_\alpha=0$. Thus we will also refer to them as *energy rays* in some instances in the remaining of this paper. The total energy is finally recovered by transporting the initial energy along the particle paths in phase space

$$\int_\mathcal{X}\varphi(\mathbf{s},\xi)W(\mathrm{d}\mathbf{s},\mathrm{d}\xi)=\sum_{\alpha=1}^{n}\int_{\mathcal{X}_\alpha}\varphi(\Phi_\tau(\mathbf{s},\xi))W_0(\mathrm{d}\mathbf{s},\mathrm{d}\xi),$$

for all continuous (test) functions φ with compact support in $T^*(\mathcal{O}\times\mathbb{R})$, introducing the sets

$$\mathcal{X}_\alpha=\{(\mathbf{s},\xi)\in\mathcal{X};\ \mathcal{H}_\alpha(\mathbf{s},\xi)=0\}. \tag{20}$$

Now the reasons why energy quantities are considered for the characterisation of oscillating solutions of wave equations may be understood as follows. At first, let us consider a function $x\mapsto u_\varepsilon(x)$ oscillating with an amplitude a about its mean $\underline{u}$ both functions being slowly varying, for example

$$u_\varepsilon(x)=\underline{u}(x)+a(x)\sin\frac{x}{\varepsilon},\quad \varepsilon\ll 1.$$

This function has no strong limit when $\varepsilon\to 0$ (in any $L^p(\mathbb{R})$, $1\le p\le +\infty$). However, if its vague limit is of interest, namely

$$\lim_{\varepsilon\to 0}\int_{\mathbb{R}}\varphi(x)(u_\varepsilon(x))^2\,\mathrm{d}x \tag{21}$$

for any continuous function φ with compact support on $\mathbb{R}$, it is found that [93]

$$\lim_{\varepsilon\to 0}\int_{\mathbb{R}}\varphi(x)(u_\varepsilon(x))^2\,\mathrm{d}x=\int_{\mathbb{R}}\varphi(x)\left((\underline{u}(x))^2+\frac{1}{2}(a(x))^2\right)\mathrm{d}x.$$

The associated "energy" admits, locally at any point, a limit given by $\underline{u}(x)^2+\frac{1}{2}a(x)^2$; see Fig. 8. It allows to "see" the deviation of oscillations with amplitude a about the mean at any point x selected by the observation function φ, whereas the weak limit of u_ε does not. At last it no longer has the oscillatory feature of the initial function, a simplification which is of high interest for experiments and simulations.

The semiclassical measure, or Wigner measure [25,26,28,33,34,82,127] can be seen as a mathematical generalisation of these concept and ideas. It is also considered to link the foregoing density W and the strongly oscillating solutions $\mathbf{u}_\varepsilon$ of (11). Let $\mathcal{S}(T^*\mathbb{R}^d)$ be the Schwartz space of $\mathcal{C}^\infty$ functions which are rapidly decreasing as well as all their derivatives (namely $\lim_{|\mathbf{z}|\to\infty}|\mathbf{z}|^k|\nabla_{\mathbf{z}}^\alpha\varphi(\mathbf{z})|=0$ $\forall k\in\mathbb{N}$, $\forall\alpha\in\mathbb{N}^{2d}$ with $\mathbf{z}\in T^*\mathbb{R}^d$) on $T^*\mathbb{R}^d\equiv\mathbb{R}^d_{\mathbf{x}}\times\mathbb{R}^d_{\mathbf{k}}$, and let φ be a so-called $n\times n$ smooth matrix observable of which coefficients are in $\mathcal{S}(T^*\mathbb{R}^d)$. For a vector field $\mathbf{u}\in L^2(\mathbb{R}^d)^n$, the space of $\mathbb{R}^n$-valued, square integrable functions equipped with the scalar product $(\mathbf{u},\mathbf{v})_{L^2}=\int_{\mathbb{R}^d}\mathbf{u}(\mathbf{x})\cdot\mathbf{v}(\mathbf{x})\,\mathrm{d}\mathbf{x}$, consider the operator

$$\Phi(\mathbf{x},\varepsilon\mathbf{D})\mathbf{u}(\mathbf{x})=\Phi(\mathbf{x},\varepsilon\mathbf{D})\left(\int_{\mathbb{R}^d}\mathrm{e}^{\mathrm{i}\mathbf{k}\cdot\mathbf{x}}\hat{\mathbf{u}}(\mathbf{k})\frac{\mathrm{d}\mathbf{k}}{(2\pi)^d}\right)=\int_{\mathbb{R}^d}\mathrm{e}^{\mathrm{i}\mathbf{k}\cdot\mathbf{x}}\varphi(\mathbf{x},\varepsilon\mathbf{k})\hat{\mathbf{u}}(\mathbf{k})\frac{\mathrm{d}\mathbf{k}}{(2\pi)^d},$$

where $\hat{\mathbf{u}}(\mathbf{k})\coloneqq\int_{\mathbb{R}^d}\mathrm{e}^{-\mathrm{i}\mathbf{k}\cdot\mathbf{x}}\mathbf{u}(\mathbf{x})\,\mathrm{d}\mathbf{x}$ stands for the Fourier transform of $\mathbf{u}(\mathbf{x})$. Then if the sequence $(\mathbf{u}_\varepsilon)$ is uniformly bounded in $L^2(\mathbb{R}^d)^n$, there exists a positive, Hermitian measure $\mathbf{W}[\mathbf{u}_\varepsilon]$ such that, up to extracting a subsequence if need be

$$\lim_{\varepsilon\to 0}(\Phi(\mathbf{x},\varepsilon\mathbf{D})\mathbf{u}_\varepsilon,\mathbf{u}_\varepsilon)_{L^2}=\mathrm{Tr}\int_{T^*\mathbb{R}^d}\varphi(\mathbf{x},\mathbf{k})\mathbf{W}[\mathbf{u}_\varepsilon](\mathrm{d}\mathbf{x},\mathrm{d}\mathbf{k}),\quad\forall\varphi\in\mathcal{S}(T^*\mathbb{R}^d)^{n,n}. \tag{22}$$

$\mathbf{W}[\mathbf{u}_\varepsilon]$ is the so-called Wigner measure of the sequence $(\mathbf{u}_\varepsilon)$ because it can also be interpreted as the weak limit of its Wigner transform $\mathbf{W}_\varepsilon(\mathbf{u}_\varepsilon,\mathbf{u}_\varepsilon)$. Indeed, if the latter is defined for temperate distributions $\mathbf{u},\mathbf{v}\in\mathcal{S}'(\mathbb{R}^d)^n$ (the dual space of $\mathcal{S}(\mathbb{R}^d)^n$) by

$$\mathbf{W}_\varepsilon(\mathbf{u},\mathbf{v})(\mathbf{x},\mathbf{k})=\frac{1}{(2\pi)^d}\int_{\mathbb{R}^d}\mathrm{e}^{\mathrm{i}\mathbf{k}\cdot\mathbf{y}}\mathbf{u}(\mathbf{x}-\varepsilon\mathbf{y})\otimes\overline{\mathbf{v}(\mathbf{x})}\,\mathrm{d}\mathbf{y}, \tag{23}$$

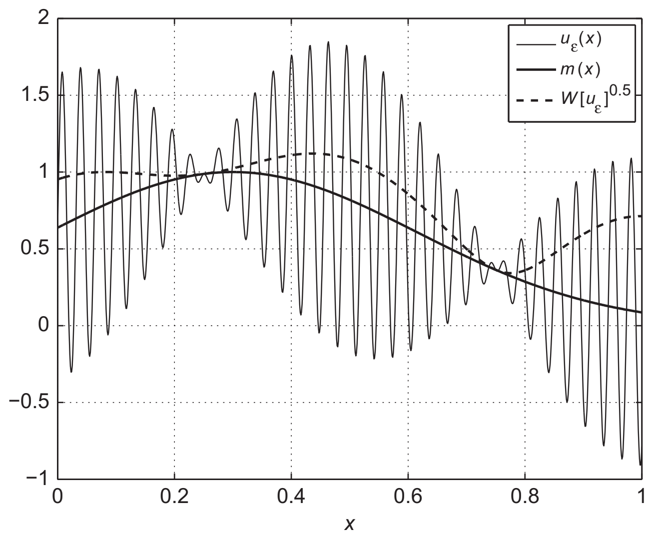

Fig. 8. The Wigner measure of a strongly oscillating sequence.

then one has

$$(\Phi(\mathbf{x},\varepsilon\mathbf{D})\mathbf{u},\mathbf{v})_{L^2}=\mathrm{Tr}\int_{T^*\mathbb{R}^d}\varphi(\mathbf{x},\mathbf{k})\mathbf{W}_\varepsilon(\mathbf{u},\mathbf{v})(\mathrm{d}\mathbf{x},\mathrm{d}\mathbf{k}).$$

Thus $\mathbf{W}[\mathbf{u}_\varepsilon]$ describes the limit energy of the sequence $(\mathbf{u}_\varepsilon)$. As in Eq. (21), the matrix function $\varphi(\mathbf{x},\mathbf{k})$ is used to select any quadratic observable or quantity of interest associated to this energy: the kinetic energy, or the free energy, or the power flow, etc. For example, the high-frequency $(\varepsilon\to 0)$ strain energy $\mathcal{V}_\varepsilon(t):=\frac{1}{2}\int_{\mathcal{O}}\mathbf{C}^e\boldsymbol{\epsilon}_\varepsilon:\boldsymbol{\epsilon}_\varepsilon\,\mathrm{d}\mathbf{x}$ in $\mathcal{O}$ (see Eq. (7)) with $\boldsymbol{\epsilon}_\varepsilon=\boldsymbol{\nabla}_{\mathbf{x}}\otimes_s\mathbf{u}_\varepsilon$ may be estimated using $\varphi(\mathbf{x},\mathbf{k})\equiv\varrho(\mathbf{x})\boldsymbol{\Gamma}(\mathbf{x},\mathbf{k})$, the acoustic tensor (13a) of the background medium

$$\lim_{\varepsilon\to 0}\mathcal{V}_\varepsilon(t)=\frac{1}{2}\int_{\mathcal{O}\times\mathbb{R}^d}\varrho(\mathbf{x})\boldsymbol{\Gamma}(\mathbf{x},\mathbf{k}):\mathbf{W}[\mathbf{u}_\varepsilon(\cdot,t)](\mathrm{d}\mathbf{x},\mathrm{d}\mathbf{k}),$$

up to some possible boundary effects on $\partial\mathcal{O}$. Similarly, the kinetic energy $\mathcal{T}_\varepsilon(t):=\frac{1}{2}\int_{\mathcal{O}}\varrho\left|\partial_t\mathbf{u}_\varepsilon\right|^2\,\mathrm{d}\mathbf{x}$ is estimated by

$$\lim_{\varepsilon\to 0}\mathcal{T}_\varepsilon(t)=\frac{1}{2}\int_{\mathcal{O}\times\mathbb{R}^d}\varrho(\mathbf{x})\mathrm{Tr}\,\mathbf{W}[\partial_t\mathbf{u}_\varepsilon(\cdot,t)](\mathrm{d}\mathbf{x},\mathrm{d}\mathbf{k}).$$

A close concept is H-measure, or microlocal defect measure [125,126,128]. The main difference is that the latter is associated to any mean-zero square integrable sequence, and it does not require any explicit scale ε as in (23). However, it is defined on the unit cosphere bundle $S^*\mathbb{R}^d\equiv\mathbb{R}^d_{\mathbf{x}}\times\mathbb{S}^{d-1}_{\hat{\mathbf{k}}}$, where $\mathbb{S}^{d-1}$ is the unit sphere of $\mathbb{R}^d$, that is it contains slightly less information than the Wigner measure since it does not depend on the norm of the wave vector.

Since the sequence $\mathbf{u}_\varepsilon$ considered above satisfies the wave equation (11), it can also be shown that: (i) its Wigner measure can be expanded on the eigenspaces of the dispersion matrix $\mathbf{H}$, *i.e.* into the sum of Wigner measures $\mathbf{W}_\alpha$ for each mode (this is the meaning of Eq. (19)); and (ii) each term in this expansion satisfies a transport equation of the form $\{\mathcal{H}_\alpha,\mathrm{Tr}\,\mathbf{W}_\alpha\}=0$. A general mathematical framework for the direct passage from a wave equation with slowly varying coefficients to a transport equation for the phase space energy density of the high-frequency solutions of the former is presented in Refs. [26,34].

3.3.3. Radiative transfer model

The transport model above can be extended to propagation media with rapidly varying parameters, at the same scale ε as the wavelength. More generally, considering random perturbations of these parameters with a correlation length ℓ_c of the same order as the wavelength, $\kappa\sim\varepsilon$, the interactions of high-frequency waves with such "high-frequency" media are depicted by a radiative transfer equation. It has the same characteristics as the transport equation (18), and the passage from a wave equation with oscillating coefficients to a radiative transfer equation is carried on using the same mathematical tools. However, only weak fluctuations have to be considered in order to observe an effective propagation pattern, otherwise the energy remains localised near its source. This passage is based on multiscale expansions and is only formal, in other words no rigorous proof exists except for some particular cases [29,32]. Nevertheless, the radiative transfer model has been widely tested and validated by its numerous applications in neutronic transfers, thermal transfers, the analyses of the optical or acoustical properties of scattering media (in astrophysics, seismology, medical imaging, multiphase flows, room acoustics, etc.), time reversal techniques, etc.; see for example [28,39,40,44,129–132]. As for the dynamics of built-up systems, considering for instance the experimental structure depicted in Fig. 1, it may be observed that the bending wavelength at the maximum experimental frequency is several fractions of the scattering mean free path, estimated by the average distance between stiffeners (about 20 cm). Thus they shall contribute, together with the junctions, to multiply scatter the waves, playing the same role as the random perturbations in the above discussion. This is also the argument developed in [132] for multiply scattered acoustic waves in fitted rooms. The radiative transfer model also allows to exhibit the diffusive behaviour of the waves at late times, as explained below in Section 3.4.

In this model, the influence of random perturbations or inclusions on the transport regime is characterised by an integral operator, the so-called *collision* operator, on the right-hand side of the Liouville equation

$$\{\mathcal{H},W\}=\mathcal{Q}(W).\tag{24}$$

Its effect is to modify the transport regime of the energy rays by multiple scattering, and possibly conversions of their polarisation modes α—see the expansion (19) of the density W

$$\mathcal{Q}(W)(\mathbf{s},\xi)=\int_{\mathbb{R}^{d+1}}\tilde{\sigma}(\mathbf{s},\xi,\xi')(W(\mathbf{s},\xi')-W(\mathbf{s},\xi))\,\mathrm{d}\xi',\tag{25}$$

where the *scattering cross-section* $\tilde{\sigma}(\mathbf{s},\xi,\xi')$ is written explicitly as a function of the power spectral densities of random perturbations [27–33]. The radiative transfer equation (24) is conservative in the sense that $\int_{\mathbb{R}^{d+1}}\mathcal{Q}(W)(\mathbf{s},\xi)\,\mathrm{d}\boldsymbol{\xi}=0$. Moreover, since $\mathrm{d}\omega/\mathrm{d}\tau=0$ and considering (19) one can remark that $\tilde{\sigma}$ necessarily has the form

$$\tilde{\sigma}(\mathbf{s},\xi,\xi')=\sum_{\alpha,\beta=1}^{n}\lambda_\alpha(\mathbf{x},\mathbf{k})\lambda_\beta(\mathbf{x},\mathbf{k}')\sigma_{\alpha\beta}(\mathbf{x},\mathbf{k},\mathbf{k}')\delta(\lambda_\beta(\mathbf{x},\mathbf{k}')-\lambda_\alpha(\mathbf{x},\mathbf{k})).\tag{26}$$

The kernel $\sigma_{\alpha\beta}(\mathbf{x},\mathbf{k},\mathbf{k}')$ gives the rate of conversion of an incident energy ray travelling in the direction of $\mathbf{k}'$ and the polarisation β into a ray travelling in the direction of $\mathbf{k}$ and the polarisation α when it is diffracted on random heterogeneities located at $\mathbf{x}$. They may be computed explicitly as functions of the auto- and cross-power spectra of the fluctuations in randomly perturbed heterogeneous media with correlation lengths comparable to the wavelengths $\kappa \sim \varepsilon$ [27–32]. The scattering processes described by Eq. (26) are such that the Hamiltonian $\mathcal{H}_\alpha$ of Eq. (16) is preserved, as requested, that is $\lambda_\alpha(\mathbf{x},\mathbf{k}) = \lambda_\beta(\mathbf{x},\mathbf{k}') := \omega \in \mathbb{R}$ is constant along the energy paths.

In transport and radiative transfer models, the relevant scale ℓ to describe the medium heterogeneities is the scattering mean free path ℓ_{sc}, which is defined as the mean square distance travelled by the rays between two successive diffractions

$$\ell_{sc} = \frac{|\mathbf{c}|}{\tilde{\Sigma}}, \quad \tilde{\Sigma}(\mathbf{s},\xi) = \int_{\mathbb{R}^{d+1}} \tilde{\sigma}(\mathbf{s},\xi,\xi')\,\mathrm{d}\xi', \tag{27}$$

with the *group velocity* $\mathbf{c} := \boldsymbol{\nabla}_\xi \mathcal{H}$ and the *total scattering cross-section* $\tilde{\Sigma}$. Scattering cross-sections of the form of Eq. (26) induce different scattering mean free paths $\ell_\alpha = |\mathbf{c}_\alpha|/\Sigma_\alpha$ for the different polarisations of which group velocities are $\mathbf{c}_\alpha := \boldsymbol{\nabla}_\xi \mathcal{H}_\alpha$. Accordingly, the total scattering cross-sections are

$$\Sigma_\alpha(\mathbf{x},\mathbf{k}) = \sum_{\beta=1}^{n} \lambda_\alpha^2(\mathbf{x},\mathbf{k}) \int_{\mathbb{R}^d} \sigma_{\alpha\beta}(\mathbf{x},\mathbf{k},\mathbf{k}')\delta(\lambda_\beta(\mathbf{x},\mathbf{k}') - \lambda_\alpha(\mathbf{x},\mathbf{k}))\,\mathrm{d}\mathbf{k}' = \sum_{\beta=1}^{n} \int_{\mathbb{S}^{d-1}} \Sigma_{\alpha\beta}(\mathbf{x},\mathbf{k};\hat{\mathbf{k}}')\,\mathrm{d}\Omega(\hat{\mathbf{k}}'), \tag{28}$$

where Ω is the uniform probability measure on the unit sphere $\mathbb{S}^{d-1}$ of which surface is $S_{d-1} = \int_{\mathbb{S}^{d-1}} \mathrm{d}\Omega$ (with the convention $\mathbb{S}^0 = \{-1,+1\}$), and $\hat{\mathbf{u}} := \mathbf{u}/|\mathbf{u}| \in \mathbb{S}^{d-1}$ for any vector $\mathbf{u} \in \mathbb{R}^d \setminus \{\mathbf{0}\}$. The variation scale $\kappa \sim \varepsilon$ of the heterogeneities is no longer observable by these models. The latter still holds relevance as long as $\lambda < \ell_{sc} < L$. As soon as $\ell_{sc} \ll L$, multiple scattering render them too much demanding in terms of accuracy and then inappropriate for measurements. Diffusion is a much more relevant model in this case. This regime is outlined in Section 3.4 below.

3.3.4. *Applications*

Transport and radiative transfer models for slender structures such as beams, plates or shells have been derived in [35,38,92,52], or for poro-viscoelastic media in [36]. The kinematics used until now for slender structures is of the first order (Timoshenko hypotheses for beams, Naghdi–Cooper hypotheses – for example – for shells). Higher-order kinematics shall be investigated in future researches. Indeed, a first-order kinematic assumption may be not refined enough to deal with short wavelengths with respect to the transverse dimensions of a beam or a shell. Regarding beams or plates for example, a Rayleigh–Lamb model in dimension one or two could be considered for application to non-destructive evaluation or structural health monitoring by ultrasonic waves. The Lamb model for plates has been studied in [133,134]. Yet the numerical simulations carried out with first-order kinematic models of beams and shells yield convincing results, both qualitatively and quantitatively [35,37,38,87,93,135]. Above all, a rational framework for the analyses and validations of the heuristic approaches outlined in Section 2, namely SEA (Section 2.3) or VCA (Section 2.4), can be proposed based on these derivations. They also allow to release some restrictive assumptions at the foundations of these formulations. As already noted in Section 2.5, the energy transport models are referred to as kinetic models in the literature, and we will stick to this terminology in the following.

3.4. *Macroscopic regime*

Now if the characteristic scale ℓ cannot be shorter than the size of a sample or a substructure, in other words if it becomes large with respect to the scattering mean free path of the medium, waves can no longer “see” its heterogeneities although they have significantly modified their amplitudes and phases. This is the macroscopic regime corresponding to the high-frequency range of vibrations of bounded structures as considered in SEA. The relevant evolution model in this case is the diffusion equation. Here the directions of the energy rays are lost because they have been diffracted, thus diverted, numerously. Corollarily, the power flows tend to vanish when the energy rays are diffusively spread within the medium by multiple scattering. Strictly speaking, the radiative transfer model in the mesoscopic regime perfectly accounts for these phenomena at long times of for short scattering mean free paths with respect to the dimensions of the sample of the propagation/observation distances L, after numerous diffractions. However, introducing a judicious rescaling and using a formal asymptotic expansion, the long-time limit of the radiative transfer equation can be exhibited: it has precisely the form of a diffusion equation. The mathematical structure of the corresponding diffusion operator depends on the shape of the random heterogeneities occupying the propagation medium, which in return define the mathematical properties of the collision operator. The diffusion regime is characterised by universal energy partition rules between different propagation modes in relation to the properties of the scattering cross-sections, in particular if the latter are isotropic

$$\tilde{\sigma}(\mathbf{s},\xi,\xi') = \frac{|\mathbf{c}|}{|\xi'|^d}\,\overline{\sigma}(\mathbf{s},|\xi|,\hat{\xi}\cdot\hat{\xi}'). \tag{29}$$

On the contrary, localisation phenomena occur for strongly anisotropic scattering cross-sections (the case of anisomeric media [136,137]). The characteristic length of these phenomena is the transport mean free path $\ell^\star$, which is the mean square distance travelled by the energy rays up to the complete loss of their directivity pattern. $\ell^\star$ is defined from the mesoscopic parameter ℓ_{sc} (the scattering mean free path) by selecting the forward component–without diffraction—of the total scattering cross-section

$$\ell^\star = \frac{|\mathbf{c}|}{\overline{\Sigma}}, \quad \overline{\Sigma}(\mathbf{s},|\xi|) = \int_{\mathbb{S}^d} \overline{\sigma}(\mathbf{s},|\xi|,\hat{\xi}\cdot\hat{\xi}')(1-\hat{\xi}\cdot\hat{\xi}')\,\mathrm{d}\Omega(\hat{\xi}').$$

This is now the relevant lengthscale ℓ for the description of the heterogeneities in the propagation medium since it is no longer conceivable to follow in detail all scattered energy paths in the diffusion regime. The energy equipartition rule invoked above reads

$$\frac{\mathcal{E}_\alpha}{\mathcal{E}_\beta} = \frac{r_\alpha}{r_\beta}\left|\frac{\lambda_\beta}{\lambda_\alpha}\right|^d, \quad \mathcal{E}_\alpha(\mathbf{s}) = \int_{\mathcal{X}_\alpha} W_\alpha(\mathbf{s},\mathrm{d}\xi), \tag{30}$$

as soon as $t > \overline{\Sigma}^{-1}$ independently of the initial condition and of the random heterogeneities ; here r_α stands for the order of multiplicity of the mode α with an energy density $\mathcal{E}_\alpha$ in $\mathcal{O}\times\mathbb{R}$, such that $\sum r_\alpha = n$. These issues constitute an active and fruitful research topic in the mathematical and physical literature; they are enlighten by numerous experiments performed among others geophysicists or acousticians (the works of Campillo, Larose, Margerin, van Tiggelen et al. [136,138–141], Fink et al. [142], Page et al. [143], Picaut et al. [144,145], Scales [146], Turner, Weaver et al. [50–53,137,147], etc.).

The diffusion macroscopic regime also arises from the mesoscopic regime when the transport parameters are randomly perturbed, for example the group velocity $\mathbf{c}$ in Eq. (18). If these perturbations have a small correlation length ℓ_c with respect to the size of the sample or the propagation distance, yet large with respect to the wavelength $\varepsilon \ll \kappa \ll 1$, the Liouville equation reduces to a Fokker–Planck equation (ray diffusion) because it still depends on the propagation direction ξ; its diffusion matrix depends on the auto-correlation function of the perturbations of the group velocity [33,62]. The Fokker–Planck equation itself reduces to a spatial diffusion equation for long propagation times/distances. Another usual case, yielding the Laplace–Beltrami diffusion regime, is the strongly forward-peaked collision operator—the so-called grazing collisions: $\tilde{\sigma}(\mathbf{s},\xi,\xi') \simeq \tilde{\Sigma}(\mathbf{x},\xi)\delta(1-\hat{\xi}\cdot\hat{\xi}')$ [148,149].

An implementation of these models for homogeneous slender structures with random perturbations is detailed in the papers [35,150,151]. The paper [68] offers a general analysis for heterogeneous thick beams or plates. The paper [36] outlines the diffusion regime for a randomly perturbed, poro-viscoelastic homogeneous background medium. At last an example of Fokker–Planck diffusion in a random thick plate is outlined in [151]. These different results partly justify the heuristic model of energy diffusion in the vibrational conductivity analogy of Section 2.4. One can in fact formally obtain a Fourier-like law (10) [68] but the derivation of the diffusive regime from the transport regime explicitly requires the consideration of the influence of random heterogeneities or boundaries (as done in the next section) in order to "mix" efficiently the different propagation directions ξ. This requirement is in basic contradiction with the derivation of Nefske and Sung [17] and subsequent research works [18–24]. All have used an original result by Rybak [152], who obtained the Fourier law and a diffusion equation starting directly from a bending or a membrane equation for a thin plate with delta-correlated random inclusions; however, these random heterogeneities within the medium have generally been discarded in the derivations. But if that medium is perfectly homogeneous, the scattering mean free path (27) is infinite and diffusion is, as a rule, never reached.

4. Application to structural dynamics

The issue of modeling dissipation effects and boundary conditions has not been addressed in the previous exposition of kinetic models. Their consideration is however a central concern for the application of the latter to structural dynamics, since it deals with damped, bounded media. Several results on dissipation phenomena in kinetic equations and boundary and interface conditions adapted to these models are outlined in Sections 4.1 and 4.2 below, respectively. This section is closed in Section 4.3 by the proposal of a transport model for the transient dynamics of elastic structures impacted by HF loads, presented as an outcome of the research works carried on until now. The continuity equation (9) in the HF limit $\varepsilon\to 0$ is ultimately given the form of a system of conservation equations for the different energy rays (the "component modes" in this very limit), supplemented with dedicated boundary and interface conditions for the reflections and transmissions of the associated power flows. This setting embodies dissipation phenomena at the boundaries should the corresponding reflection/transmission operators be non-conservative.

4.1. Damping modeling

The influence of memory effects in viscoelastic materials can be considered starting from a stress-strain constitutive relation of the form (6). It has been shown [91] that the memory term (the integral term) has no influence on the kinetic equations obtained for the evolution of the high-frequency energy density. As for poro-viscoelastic media, the solid and fluid phases memory effects (the viscosity and compressibility are time-dependent) are also ineffective on the transport regime. The analysis in [36] is done along the same lines as in [91]. These results are in basic agreement with the current practical observations: viscoelastic materials tend to respond as purely elastic materials when they are forced by high-frequency harmonic motions (see for example [153, p. 22–23]). This conclusion does not mean however that no internal dissipation occurs in the high-frequency range, it rather indicates that the viscous model (6) may be irrelevant in the transport regime. The radiative transfer equation (24) embodies dissipative phenomena if for example its collision operator $\mathcal{Q}$ has the form

$$\mathcal{Q}(W)=\mathcal{Q}^{(0)}(W)-\eta W, \quad \int_{\mathbb{R}^{d+1}}\mathcal{Q}(W)(\mathbf{s},\xi)\,\mathrm{d}\xi<0, \tag{31}$$

where $\mathcal{Q}^{(0)}$ is the conservative part and $\eta>0$ is a loss factor (in s $^{-1}$) which describes the absorption rate of energy "particles" in the collision process. $\mathcal{Q}$ is said to be subcritical in this case. The solution W of the radiative transfer equation (24) is in this case

$$W(\mathbf{s}(\tau),\xi(\tau))=\mathrm{e}^{-\eta\tau}W^{(0)}(\mathbf{s}(\tau),\xi(\tau)), \tag{32}$$

where $W^{(0)}$ is the solution – if any – without absorption. Thus the model of Eq. (31) is representative of an energy density decreasing exponentially along its paths in phase space.

The issue of establishing the constitutive equation yielding a given form of subcritical collisions and the reverse then arises. In [89,90] the authors have shown that for a scalar wave equation (4) and a Kelvin–Voigt constitutive equation, $\boldsymbol{\sigma}=\mathbf{C}^e\boldsymbol{\epsilon}+\mathbf{C}^\nu\dot{\boldsymbol{\epsilon}}$, the loss factor is

$$\eta(\mathbf{x},\mathbf{k})=2\frac{\mathbf{C}^e(\mathbf{x})\mathbf{k}\cdot\mathbf{k}}{\mathbf{C}^\nu(\mathbf{x})\mathbf{k}\cdot\mathbf{k}},$$

that is to say η is twice the rate of relaxation, as expected. For more general forms of subcritical collision operators, this issue is still rather open and raises difficult questions. It is the subject of ongoing researches in the applied mathematics community.

4.2. Boundary and interface conditions

Boundary conditions for quadratic quantities such as the vibrational energy density shall be constructed on the basis of the boundary conditions applied to the underlying displacement and stress fields, for example Dirichlet or Neumann boundary conditions, or interface jump conditions between substructures (assuming that the thickness of that interface is much smaller than the wavelength) [82–86]. They basically translate into reflection/transmission operators for the energy fluxes. The consideration of boundary conditions in kinetic models raises some significant theoretical difficulties related to the polarisation and the possible conversion of waves in elastodynamics, as well as in electromagnetism, and to the critical angles of incidence arising for either transmission or reflection problems [88]. Indeed, the energy "density" W having its support in the sets $\mathcal{X}_\alpha$ by Eq. (19), the condition $\mathcal{H}_\alpha(\mathbf{s},\xi)=0$ has yet to be satisfied on the boundary $\partial\mathcal{O}\times\mathbb{R}$, or on an interface $\Sigma_D\subset\mathcal{O}\times\mathbb{R}$ oriented by its normal $(\mathbf{n},n_t)$. Commonly for acoustic or elastic waves in three-dimensional media and slender structures, the eigenvalues λ_α^2 of the Christoffel tensor $\boldsymbol{\Gamma}$ have the form $\lambda_\alpha(\mathbf{x},\mathbf{k})=c_\alpha(\mathbf{x})|\mathbf{k}|$ where $c_\alpha>0$ is the phase velocity of the αth mode. Thus this very condition reads

$$\mathcal{H}_\alpha(\mathbf{s},\xi)=\varrho(\mathbf{x})c_\alpha^2(\mathbf{x})k_\mathrm{n}^2+\mathcal{H}_\alpha(\mathbf{s},\mathbf{k}^\star,\omega)=0, \tag{33}$$

isolating the normal component $k_\mathrm{n}=\mathbf{k}\cdot\hat{\mathbf{n}}$ of the wave vector, while $\mathbf{k}^\star=(\mathbf{I}_d-\hat{\mathbf{n}}\otimes\hat{\mathbf{n}})\mathbf{k}$ is its tangential component. Eq. (33) is a simple quadratic equation for k_n, parameterised by $\mathbf{k}^\star$. It has real or purely imaginary solutions if $\mathcal{H}_\alpha(\mathbf{s},\mathbf{k}^\star,\omega)<0$ or $\mathcal{H}_\alpha(\mathbf{s},\mathbf{k}^\star,\omega)>0$, respectively, since $\varrho(\mathbf{x})c_\alpha^2(\mathbf{x})>0$. Thus the cotangent bundle to the boundary $T^*\partial\mathcal{O}\times\mathbb{R}$ splits as $T^*\partial\mathcal{O}\times\mathbb{R}=\mathcal{H}_\alpha\cup\mathcal{E}_\alpha\cup\mathcal{G}_\alpha$ (for all modes α, see for example [84]), where the different diffraction regions on $\partial\mathcal{O}\times\mathbb{R}$ are defined by

$$\begin{aligned}\mathcal{H}_\alpha&=\{(\mathbf{s},\mathbf{k}^\star,\omega)\in T^*\partial\mathcal{O}\times\mathbb{R};\ \mathcal{H}_\alpha(\mathbf{s},\mathbf{k}^\star,\omega)<0\}\quad\text{(hyperbolic)},\\ \mathcal{E}_\alpha&=\{(\mathbf{s},\mathbf{k}^\star,\omega)\in T^*\partial\mathcal{O}\times\mathbb{R};\ \mathcal{H}_\alpha(\mathbf{s},\mathbf{k}^\star,\omega)>0\}\quad\text{(elliptic)},\\ \mathcal{G}_\alpha&=\{(\mathbf{s},\mathbf{k}^\star,\omega)\in T^*\partial\mathcal{O}\times\mathbb{R};\ \mathcal{H}_\alpha(\mathbf{s},\mathbf{k}^\star,\omega)=0\}\quad\text{(tangent)}.\end{aligned}$$

The first one corresponds to the transverse rays (below critical incidence) for which k_n is real, the second one corresponds to the totally reflected rays (above critical incidence) for which k_n is purely imaginary, and the third one corresponds to the tangent rays (critical or tangential incidence) for which $k_\mathrm{n}=0$ in the local frame of the tangent plane to the boundary at $\mathbf{s}$. As for an interface, these definitions have to be extended on both sides since the acoustic tensor, and thus its eigenvalues, are *a priori* different; $T^*\Sigma_D$ is then the union of all these regions [84], but the latter are not necessarily disjoint. Similarly to the collisions of the rays on random heterogeneities of the medium detailed in Section 3.3.3, mode conversions $\alpha\leftrightarrow\beta$ in $\mathcal{H}_\alpha\cap\mathcal{H}_\beta$ or $\mathcal{H}_\alpha\cap\mathcal{E}_\beta$ are driven by the condition $\mathrm{d}\omega/\mathrm{d}\tau=0$ derived from Eq. (16). It enforces the conservation of the Hamiltonian, that is $\lambda_\alpha(\mathbf{x}(\tau_0^-),\mathbf{k}(\tau_0^-))=\lambda_\beta(\mathbf{x}(\tau_0^+),\mathbf{k}(\tau_0^+))$ on any discontinuity front Σ_D of the density W given by $\Sigma_D=\{(\mathbf{s},\xi)\in T^*(\mathcal{O}\times\mathbb{R});\ \mathcal{S}(\mathbf{s},\xi)=0\}$, where therefore $\tau_0>0$ is such that $\mathcal{S}(\mathbf{s}(\tau_0),\xi(\tau_0))=0$. Besides, the Rankine–Hugoniot condition on $\Sigma_D\cap(\bigcup_{\alpha=1}^n\mathcal{X}_\alpha)$ is for the transport or radiative transfer equation

$$[\![\{\mathcal{H},\mathcal{S}\}W]\!]=0,$$

where $[\![f]\!]=f(\tau_0^+)-f(\tau_0^-)$ stands for the jump of f on the front. This condition expresses the conservation of the normal energy flux on the discontinuity, but it does not describe *a priori* how the energy density is distributed among the different modes by the reflections and transmissions on an interface (a junction) Σ_D independent of time for example. However, solving a Riemann problem at that interface gives a decomposition of the normal energy flux into rays moving forward and backward from it, including the effects of reflection and transmission. Thus the power flow reflection and transmission coefficients for the transport problem may be derived and understood in terms of some particular Riemann solutions [73,154]. It may be observed in addition that since W_β is necessarily zero in a neighbourhood of $\mathcal{E}_\beta$ (away from $T^*\Sigma_D$) because its support is in $\mathcal{X}_\beta$, its trace on $T^*\Sigma_D$ is zero as well, thus $W_\beta \equiv 0$ by the mode conversion $\alpha \to \beta$ within $\mathcal{H}_\alpha \cap \mathcal{E}_\beta$.

The strategy adopted in [155] (see also [156,157]) consists in constructing an approximate solution of the wave equation (11) for which an explicit expression of its energy density is known, since the latter is also the one of the exact high-frequency solution. This approximation is chosen for instance as an infinite sum of Gaussian beams weighted by the Fourier–Bros–Iagolnitzer (FBI, see for example [158, Chapter 3]) transform of the initial conditions, provided that their support is disconnected from the boundary. The whole analysis is however restricted to transverse reflections solely (in $\mathcal{H}_P \cap \mathcal{H}_S$) for an isotropic elastic medium. For slender structures such as thick beams or plates, the power flow reflection/transmission coefficients at boundaries or interfaces can be computed directly on the basis of the dispersion properties of the constitutive and balance equations for these systems [38,92]. For Mindlin–Reissner–Uflyand's thick plate kinematic model, the energy density is split into three propagative modes $\alpha \in \{T,S,P\}$ of multiplicity 1, 2 and 2 respectively (thus $n=5$), with $\lambda_\alpha(\mathbf{x},\mathbf{k}) = c_\alpha(\mathbf{x})|\mathbf{k}|$ such that $c_T < c_S < c_P$. Polarisation T corresponds to transverse shear and the polarisations S, P correspond to bending and in-plane vibrational energies in the mean surface of the plate. The normal power flow reflection/transmission coefficients at the junction of plates and at a fixed or a free boundary are derived for the hyperbolic regions and the hyperbolic–elliptic region as functions of the angle between the plates.

In these various works the energy possibly guided by interfaces and boundaries in the so-called glancing regions $\mathcal{G}_\alpha$ is actually ignored, not because it is negligible, but because a theoretical model lacks at present for the elastic case. However, it may be observed in practice that a significant part of the energy flows is likely to travel along the interfaces, junctions or stiffeners: it is therefore necessary to take it into account in the analysis and models. Ref. [84] develops a rigorous, yet still partial, mathematical analysis of the Wigner measure of acoustic waves in the glancing region. In [159] a finite difference numerical scheme in phase space is proposed in order to deal with all diffractive regions at a curved interface between two acoustic media, including the glancing one. However, the diffraction models implemented in this approach do not seem to be coherent with a kinetic model for energetic quantities.

The influence of irregularities or non-smoothness of the boundary and interfaces, such as corners or wedges, is also ignored. Keller's geometrical theory of diffraction (GTD) [160] has been introduced as a corrector in ray methods to account for these effects. The theory should be extended to kinetic models (18)—a perspective which seems today out of reach owing to its deep theoretical difficulty. Ref. [161] is a first attempt, to our knowledge, to construct a numerical method for the simulation of high-frequency acoustic waves diffraction by a half-plane, relying on GTD.

4.3. Outcome: multigroup radiative transfer equations

From the previous considerations and the results outlined in Section 3.3, a kinetic model of transient structural dynamics for high-frequency loads is derived as follows. To begin with, the Liouville equation (18) or the radiative transfer equation (24) is written in the form of a first-order system of so-called multigroup transport equations, invoking Eq. (19) to single out the rays $\tau \mapsto \mathbf{x}_\alpha(\tau)$ and the densities $\tau \mapsto W_\alpha(\tau)$ of the different modes α. Let $E = \{1,2,\ldots,n\}$ be the discrete set of these propagative modes, corresponding to the different groups in the terminology of neutron or thermal transfers [131]. Also let $\mathbf{z} = (\mathbf{x},\mathbf{k}) \in T^*\mathcal{O} := \mathcal{O} \times \mathbb{R}^d$, and $\mathbf{w} = (W_1,W_2,\ldots,W_n)^\mathsf{T} \in (\mathbb{R}_+)^n$; the linear flux operator $\mathcal{F}$ is defined by

$$\mathcal{F}(\mathbf{w}) = (W_1\mathbf{F}_1, W_2\mathbf{F}_2, \ldots W_n\mathbf{F}_n)^\mathsf{T}, \quad \mathbf{F}_\alpha(\mathbf{z}) = \begin{pmatrix} \boldsymbol{\nabla}_\mathbf{k}\lambda_\alpha \\ -\boldsymbol{\nabla}_\mathbf{x}\lambda_\alpha \end{pmatrix},$$

where $\alpha \in E$ (thus $\mathcal{F}(\mathbf{w})$ is a $n \times 2d$ real matrix). Typically in acoustics, elastodynamics or for the vibrations of slender structures, the group velocities are $\mathbf{c}_\alpha(\mathbf{z}) := \boldsymbol{\nabla}_\mathbf{k}\lambda_\alpha(\mathbf{z}) = c_\alpha(\mathbf{x})\hat{\mathbf{k}}$ where $c_\alpha > 0$ is the phase velocity of waves for the polarisation mode α (see also Section 4.2) and $\hat{\mathbf{k}} \in \mathbb{S}^{d-1}$. Hence the multigroup radiative transfer equations (MRTE) read

$$\partial_t \mathbf{w} + \mathbf{Div}_\mathbf{z}\mathcal{F}(\mathbf{w}) = \mathcal{Q}(\mathbf{w}). \tag{34}$$

Indeed, it is deduced from Eq. (16) that $\mathrm{d}t/\mathrm{d}\tau = -2\omega\varrho(\mathbf{x})$ where ω is a non-zero constant; the radiative transfer equation (24) has the form (34) up to the multiplicative constant ω, and the latter does not come up in the subsequent analyses. The right-hand side is the integral collision operator $\mathcal{Q} \in L^1_\mathbf{k}(\mathbb{R}^d)$ of Eq. (24), which is assumed to be independent of time from now on. It can be seen as a linear combination of source and absorption contributions. However, it remains globally conservative in the sense that $\int_{\mathbb{R}^d} \mathcal{Q}(\mathbf{w})(\mathbf{x},\mathbf{k})\,\mathrm{d}\mathbf{k} = \mathbf{0}$, $\forall \mathbf{x} \in \mathcal{O}$. The model (34) is supplemented by the initial condition $\mathbf{w}(\mathbf{z},0) = \mathbf{w}_0(\mathbf{z})$ such that $\mathbf{w}_0(\mathrm{d}\mathbf{x},\{\mathbf{k}=\mathbf{0}\}) = \mathbf{0}$. Hence, the radiative transfer equation (24) has the form of a system of linear conservation equations (34) for the n energy densities W_α, $\alpha \in E$, lying in a domain $\mathcal{W} \subset (\mathbb{R}_+)^n$; it can be compared to Eq. (9) of Section 2. One may first observe that the constant function $\mathbf{w}(\mathbf{z},t) = \overline{\mathbf{w}}$ is a trivial solution of (34). In return for a classical hyperbolicity assumption, that is to say the (dispersion) matrix $\boldsymbol{\Gamma}(\overline{\mathbf{w}},\boldsymbol{\zeta}) := \boldsymbol{\zeta} \cdot \mathbf{D}_\mathbf{w}\mathcal{F}(\overline{\mathbf{w}})$ is diagonalisable on $\mathbb{R}$ uniformly with respect to $(\overline{\mathbf{w}},\boldsymbol{\zeta}) \in \mathcal{W} \times \mathbb{R}^{2d}\backslash\{\mathbf{0}\}$, the Cauchy problem associated to Eq. (34) is well-posed in $L^2(T^*\mathcal{O})^n$ [162, Chapter 2]. The Maxwell equations in electromagnetism, the Euler equations for non-viscous gas dynamics, the Navier–Stokes equations for viscous fluid dynamics, or the elastic wave equations can all be written in the form (34)—possibly with a nonlinear flux operator. Most of these systems satisfy in addition a symmetrisability property [162, Chapter 13], which implies that the Cauchy problem admits a regular solution locally in time, but the latter usually blows up after a finite time. Therefore, weak solutions have to be considered, that is in the sense of distributions, which may exhibit discontinuities. Piecewise regular solutions are the easiest ones to describe: they are continuous out of an at most countable set of discontinuity fronts (shocks) corresponding to hypersurfaces $\Sigma(\mathbf{z},t)=0$ of $T^*\mathcal{O}\times\mathbb{R}_t^+$ where they satisfy in addition the Rankine–Hugoniot jump condition

$$[\![\,\mathbf{w}\,]\!]n_t+[\![\,\mathcal{F}(\mathbf{w})\,]\!]\mathbf{n}=\mathbf{0}, \tag{35}$$

$[\![f]\!]$ standing for the jump of f on a discontinuity front oriented by its normal vector $(\mathbf{n},n_t)$.

For applications to structural dynamics, the previous system is subsequently considered in a bounded domain $\mathcal{D}\subseteq\mathcal{O}$. Boundary conditions on $\Gamma^{\pm}(\partial\mathcal{D})$ must be added to Eq. (34) in order to formulate a well-posed initial-boundary value problem [63,64]. The inward and outward boundaries are defined by

$$\Gamma^{\pm}(\mathcal{M}) := \{(\mathbf{x},\mathbf{k})\in\mathcal{M}\times\mathbb{R}^d;\;\; \pm\hat{\mathbf{k}}\cdot\hat{\mathbf{n}}(\mathbf{x})>0\}$$

for a smooth manifold $\mathcal{M}$ of $\mathbb{R}^d$ oriented by its unit outward normal $\hat{\mathbf{n}}$. These boundary conditions read $\mathcal{F}(\mathbf{w})\hat{\mathbf{n}}+\mathbf{f}=\mathbf{0}$ on $\Gamma^-(\partial\mathcal{D})$, where $\mathbf{f}(\mathbf{z},t)$ is a given function. Invoking the results presented in the previous section, the outward energy fluxes on $\Gamma^+(\partial\mathcal{D})$ are either totally or partially reflected by the boundary, so they take part in the inward fluxes. In this respect, the boundary conditions on $\Gamma^-(\partial\mathcal{D})$ can be generalised as

$$\mathcal{F}(\mathbf{w})\hat{\mathbf{n}}+\mathcal{R}_{\mathbf{x}}(\mathbf{w})+\mathbf{f}=\mathbf{0},\quad \mathbf{z}\in\Gamma^-(\partial\mathcal{D}),\;\; t>0, \tag{36}$$

where $\mathcal{R}_{\mathbf{x}}$ is a bounded linear operator from $\Gamma^+(\partial\mathcal{D})$ (outward boundary) to $\Gamma^-(\partial\mathcal{D})$ and $\hat{\mathbf{n}}$ is the unit outward normal to $\partial\mathcal{D}$. It depends locally on the geometry of $\partial\mathcal{D}$, and describes how the outward normal fluxes $\mathcal{F}(\mathbf{w})\hat{\mathbf{n}}$, $\hat{\mathbf{k}}\cdot\hat{\mathbf{n}}>0$, are reflected into inward normal fluxes $-\mathcal{F}(\mathbf{w})\hat{\mathbf{n}}$, $\hat{\mathbf{k}}\cdot\hat{\mathbf{n}}<0$, with potential mode conversions. Here the glancing sets $\hat{\mathbf{k}}\cdot\hat{\mathbf{n}}=0$ are ignored, see Section 4.2. As already mentioned above, a rigorous derivation of this operator is proposed in [155] for the acoustic and elastic cases with Dirichlet or Neumann boundary conditions; this analysis is carried out in [34] using semiclassical (Wigner) measures. Integrating Eq. (34) on $T^*\mathcal{D}\times E=\mathcal{D}\times\mathbb{R}^d\times E$ and using Ostrogradski's formula, yields the energy balance

$$\sum_{\alpha\in E}\partial_t\int_{T^*\mathcal{D}}W_\alpha(\mathrm{d}\mathbf{z},t)=\sum_{\alpha\in E}\int_{\Gamma^-(\partial\mathcal{D})}f_\alpha(\mathbf{z},t)\,\mathrm{d}\gamma(\mathbf{z}), \tag{37}$$

where $\mathrm{d}\gamma$ is the natural measure of $\partial\mathcal{D}\times\mathbb{R}^d$. The reflection operator has however to be non-dissipative, that is

$$\sum_{\alpha\in E}\int_{\Gamma^-(\partial\mathcal{D})}(\mathcal{R}_{\mathbf{x}}(\mathbf{v}))_\alpha\mathrm{d}\gamma(\mathbf{z})=\sum_{\alpha\in E}\int_{\Gamma^+(\partial\mathcal{D})}(\mathcal{F}(\mathbf{v})\hat{\mathbf{n}})_\alpha\mathrm{d}\gamma(\mathbf{z}),\quad \forall\mathbf{v}\in\mathcal{W}.$$

It is dissipative if the left-hand side is strictly lower to the right-hand side.

Finally, it is assume that the physical domain $\mathcal{D}$ is partitioned into non-overlapping subdomains with different group velocities, for example structural junctions of beams or shells. Their interface $\Sigma_D\subset\overline{\mathcal{D}}$ is a smooth, bounded manifold of codimension 1 oriented by its unit normal $\hat{\mathbf{n}}_D$, ignoring edges and corners at a first glance. More specifically, Σ_D is any interface between these subdomains or between a subdomain and the exterior of $\mathcal{D}$ and consequently, it may also be a part or the whole of $\partial\mathcal{D}$. The Rankine–Hugoniot condition (35) satisfied by the piecewise continuous (weak) solutions $\mathbf{w}$ – if any – is

$$\sum_{\alpha\in E}[\![\,(\hat{\mathbf{n}}_D\cdot\mathbf{c}_\alpha)W_\alpha\,]\!]=0.$$

It states that the total normal energy flux density remains continuous across that front; in return the specific intensities (W_α) themselves have no reason to be continuous. If the phase velocities $c_\alpha=|\mathbf{c}_\alpha|$ are continuous across Σ_D, the continuity of the normal energy flux on Σ_D is ensured by the conditions $\mathcal{F}(\mathbf{w}^+)\hat{\mathbf{n}}_D=\mathcal{F}(\mathbf{w}^-)\hat{\mathbf{n}}_D$ if $\hat{\mathbf{k}}\cdot\hat{\mathbf{n}}_D>0$ (forward flux) and $-\mathcal{F}(\mathbf{w}^-)\hat{\mathbf{n}}_D=-\mathcal{F}(\mathbf{w}^+)\hat{\mathbf{n}}_D$ if $\hat{\mathbf{k}}\cdot\hat{\mathbf{n}}_D<0$ (backward flux), where $\mathbf{w}^{\pm}(\mathbf{x},\mathbf{k},t)=\lim_{h\downarrow 0}\mathbf{w}(\mathbf{x}\pm h\hat{\mathbf{n}}_D,\mathbf{k},t)$ for a.e. $\mathbf{x}\in\Sigma_D$, and $\mathbf{w}^+(\mathbf{x},\mathbf{k},t)=\mathbf{0}$ whenever $\mathbf{x}\in\partial\mathcal{D}$; likewise, $c_\alpha^{\pm}(\mathbf{x})=\lim_{h\downarrow 0}c_\alpha(\mathbf{x}\pm h\hat{\mathbf{n}}_D)$ a.e. $\mathbf{x}\in\Sigma_D$. If however the velocities c_α are discontinuous across Σ_D, such that $c_\alpha^+\neq c_\alpha^-$, these expressions may be generalised to the form

$$\begin{aligned}\mathcal{F}(\mathbf{w}^+)\hat{\mathbf{n}}_D&=\mathcal{R}_D^+(\mathbf{w}^+)+\mathcal{T}_D^-(\mathbf{w}^-),\quad \mathbf{z}\in\Gamma^+(\Sigma_D),\\ -\mathcal{F}(\mathbf{w}^-)\hat{\mathbf{n}}_D&=\mathcal{R}_D^-(\mathbf{w}^-)+\mathcal{T}_D^+(\mathbf{w}^+),\quad \mathbf{z}\in\Gamma^-(\Sigma_D),\end{aligned} \tag{38}$$

$t>0$, where $\mathcal{R}_D^{\pm}:\Gamma^{\mp}(\Sigma_D)\to\Gamma^{\pm}(\Sigma_D)$ and $\mathcal{T}_D^{\mp}:\Gamma^{\pm}(\Sigma_D)\to\Gamma^{\pm}(\Sigma_D)$ are the upstream (−) and downstream (+) reflection and transmission operators, respectively, of the interface Σ_D. These relations show how the inward and outward normal energy fluxes are distributed on the interface: the outward flows are the sum of reflected and transmitted flows, taking into account possible mode conversions in E, on either side of that interface; see Fig. 9. These reflection and transmission operators satisfy

$$\sum_{\alpha\in E}\left(\int_{\Gamma^{\pm}(\Sigma_D)}(\mathcal{T}_D^{\mp}(\mathbf{v}))_\alpha\,\mathrm{d}\gamma(\mathbf{z})+\int_{\Gamma^{\mp}(\Sigma_D)}(\mathcal{R}_D^{\mp}(\mathbf{v}))_\alpha\,\mathrm{d}\gamma(\mathbf{z})\right)=\pm\sum_{\alpha\in E}\int_{\Gamma^{\pm}(\Sigma_D)}(\mathcal{F}(\mathbf{v})\hat{\mathbf{n}}_D)_\alpha\,\mathrm{d}\gamma(\mathbf{z}),\quad \forall\mathbf{v}\in\mathcal{W},$$

or a strict inequality if the interface is dissipative. Their explicit derivation for junctions of thick beams or shells is outlined in [38]. The consideration of absorbing boundaries and/or interfaces is a subject of ongoing research in the dedicated literature; see *e.g.* [163–167]. It should be noted to conclude on this topic that the above reflection/transmission operators need not be diffusive in order to yield diffuse waves and rays within the medium; see for example [135] and many applications in room acoustics [117,132], among others. The onset of a diffusive regime at late times is prompted by both the multiple scattering of energy rays on random heterogeneities, and to a greater extent on boundaries and interfaces as seen in [35,87,92,93,135].

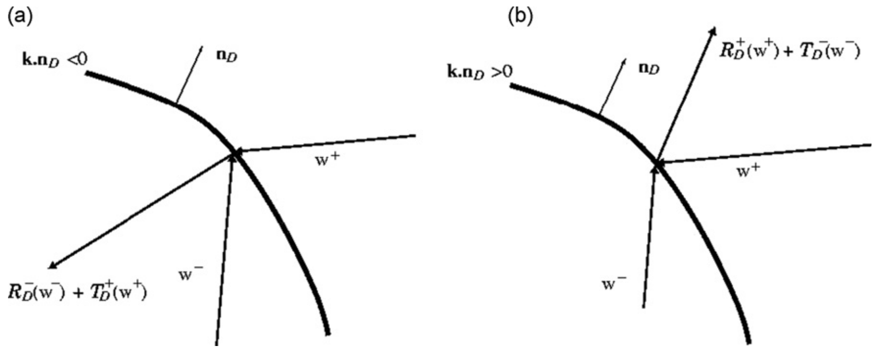

Fig. 9. Reflected and transmitted fluxes on an interface Σ_D: (a) backward flux $\hat{\mathbf{k}} \cdot \hat{\mathbf{n}}_D < 0$, (b) forward flux $\hat{\mathbf{k}} \cdot \hat{\mathbf{n}}_D > 0$.

4.4. Numerical methods and example

This presentation of kinetic models for structural dynamics is concluded by a brief overview of numerical issues and an example showing their consistency with the engineering approaches. Transport and diffusion equations can be solved numerically by various methods [65,73,131,168]. Two classes of numerical methods, namely Monte-Carlo methods [122] and nodal/spectral discontinuous finite element methods [169–173], have been particularly considered. Indeed, numerical methods with low numerical dispersion and dissipation errors are needed to perform long-time simulations of the MRTE (34) in order to reach its diffusion limit, if any. The aforementioned methods both have these properties. Their implementation for the MRTE (34) is described in more detail in [93,135].

4.4.1. Monte-Carlo methods

Monte-Carlo methods are easy to implement since they are based on a physical interpretation of the radiative transfer equations. They also allow to compute local solutions, a decisive advantage over energetic methods (such as finite elements) for large-scale computations. They converge slowly, but it is always possible to control their accuracy. Their main drawback is their lack of versatility for complex geometries as typically encountered in structural dynamics. The Monte-Carlo method for solving a transport partial differential equation or a radiative transfer integro-differential equation is based on its interpretation as a Fokker–Planck equation for the marginal probability density of an underlying homogeneous Markov chain having its values in $T^*\mathbb{R}^d \times E$ [122]. The latter corresponds to the position and velocity of particles with different polarisations in E, and the algorithm consists in simulating trajectories of these particles and subsequently averaging over these paths. It extends to a bounded domain $\mathcal{D}$ by constructing those trajectories from the broken bicharacteristic flow within $T^*\mathcal{D}$, assuming that its intersection with the glancing set remains empty for all times. This is practically done by reflecting and transmitting the energy rays in $\mathcal{D}$ according to Snell–Descartes laws with all possible mode conversions in E. If the energy densities W_α are matrices as in [26,28,34,36], this algorithm is less standard and has to be modified as done in [174] for matrix transport equations. Note that it can also be adapted to the consideration of a subcritical collision operator of the form (31) [122]. Numerical examples using the Monte-Carlo method for the simulation of energy transfers in slender structures or the Earth crust may be found in [35,38,93,135,175,176]. The method applies to a diffusion equation as well, the underlying stochastic process being an homogeneous diffusion solution of a stochastic differential equation in the Itô sense. This representation has been used in [151] for the numerical resolution of a ray diffusion equation, describing the evolution of the bending energy density in a thick plate with a randomly perturbed group velocity.

4.4.2. Discontinuous finite elements

Finite element methods are, on the contrary, much more flexible and can be applied to truly complex geometries. Among these methods, the discontinuous "Galerkin" (DG) finite element method has originally been introduced by Reed and Hill [169] and Lesaint and Raviart [170] to compute neutronic transfers; it is therefore adapted to the integration of radiative transfer equations. In the DG method, boundary fluxes at the edges or faces of the elements maintain the consistency of the numerical solution with the continuous, non-discretised kinetic equations, which are otherwise discretised on each element without any continuity relation between elements. The numerical fluxes are thus constructed in order to get to satisfy the boundary and interface conditions (36) and (38). As shown in [93,135], the DG formulation embodies, in its various forms, the penalisation method applied to the boundary fluxes of subdomains (or elements), or the ultra-weak variational formulation originally proposed by Després [177]. Depending on the choice of the trial space and the numerical fluxes, the latter is also a generic formulation for wave-based methods, Trefftz methods, least-square methods, etc., as discussed extensively in, *e.g.*, Refs. [178–180]. Numerical examples using the DG method for the simulation of energy transfers in slender structures may be found in [37,38,87,93,135].

4.4.3. A numerical example and connections with SEA

This example is taken from [93]. It deals with an assembly of two thick cylindrical shells, the first one with a constant radius (shell #1) and the second one with a variable radius (shell #2). Both cylinders have the same length L which is also twice the constant radius of the first shell. Five types of energy rays propagate in such a system: a pure shear transverse mode at the velocities c_{Tj} ($j=1$ or 2 for either shell #1 or shell #2) denoted by subscript $\alpha=T$, and coupled longitudinal and transverse modes $\alpha=P$ and $\alpha=S$ each of multiplicity 2 corresponding to either in-plane (subscript n) or bending energies (subscript b), at the velocities c_{Pj} and c_{Sj} respectively; see [35] for the details on the derivation. Then the energy density evolution in this system is depicted by five coupled radiative transfer equations (34) with $n=5$ – since $E=\{T,P\mathrm{b},P\mathrm{n},S\mathrm{b},S\mathrm{n}\}$ – and $d=2$, where the coupling is ensured by the left $(-)$ and right $(+)$ power reflection and transmission operators $\mathcal{R}_D^{\pm}$, $\mathcal{T}_D^{\pm}$ (see Eq. (38)) of the shell junction. Their expressions as functions of the angle between the shells, their mechanical and geometrical characteristics, and the incident wave vector on the junction are derived in [38]. Neumann boundary conditions are considered at the extremities of the shells; the corresponding power reflection coefficients have been derived in [38] as well. Here it has been shown that the shear mode is uncoupled to the in-plane and bending modes by boundary reflections; however, the longitudinal P and transverse S modes are coupled. Otherwise all these modes are fully coupled by the junction.

Both shells have the same thicknesses and average Young's modulii and densities, their Poisson's ratio being $\nu_1=0.3$ and $\nu_2=0.2$ respectively. However, Young's modulus and density of shell #1 are randomly perturbed such that the energy transport regime in it is described by a radiative transfer equation. The corresponding scattering cross-sections of Eq. (28) are derived in [35,93] for statistically isotropic perturbations. The coupled radiative transfer equations (34) are solved in this example by the direct Monte-Carlo method briefly exposed in Section 4.4.1. About 10^6 sample paths have been simulated for the estimation of their solution, as in [93]. The initial condition is an isotropic shear impact $W_T^0(\mathbf{x},\mathbf{k})=\delta(\mathbf{x}-\mathbf{x}_0)$, independently of $\mathbf{k}$, all other modes being initially unloaded. A Gaussian model of correlation of random inhomogeneities is used for shell #1, while shell #2 remains unperturbed; no damping is accounted for either in both shells. Fig. 10 displays the ratios $\mathcal{E}_{Pj}^h(t)/\mathcal{E}_{Sj}^h(t)$ of the total bending/longitudinal and transverse energies within each shell, respectively

$$\mathcal{E}_{Pj}^h(t)=\sum_{\alpha=P\mathrm{n},P\mathrm{b}}\int_{\mathcal{D}_j\times\mathbb{S}^1}W_\alpha^h(\mathbf{x},\hat{\mathbf{k}},t)\,\mathrm{d}\mathbf{x}\,\mathrm{d}\Omega(\hat{\mathbf{k}}),\quad j=1,2,$$

and a similar expression for $\mathcal{E}_{Sj}^h(t)$; $W_\alpha^h(\mathbf{x},\hat{\mathbf{k}},t)$ being the numerical solutions of Eq. (34) and $\mathcal{D}_j$ the mean surface of shell #j. The time scale is $T=L/c_{T2}$. For an isotropic collision operator, the diffusive limit at large times of $\mathcal{E}_{Pj}^h(t)/\mathcal{E}_{Sj}^h(t)$ has been shown to be the ratio c_{Sj}/c_{Pj} to the power of the physical dimension $d=2$, independently of the source type and scatterers [28,68]. This universal equilibration rule is recovered by the Monte-Carlo scheme, as seen in Fig. 10.

It is noteworthy that these time-domain simulations meet with the frequency domain results of an SEA computation (see Section 2.3). The modal densities $n_\alpha(\omega)$ for bending and shear eigenmodes of an isotropic plate of area $\mathcal{A}$ and perimeter $\mathcal{P}$ are given by [11]

$$n_\alpha(\omega)=\frac{\mathcal{A}\omega}{2\pi c_\alpha^2}+\mathcal{P}\gamma(\omega),\quad \alpha=T,P,S, \tag{39}$$

where $\omega\mapsto\gamma(\omega)$ depends on the boundary conditions but is such that $\lim_{\omega\to+\infty}\gamma(\omega)=0$. Sub-structuring each shell in the SEA sense into three subsystems, one for shear (subsystem #1) and two for bending motions corresponding to the

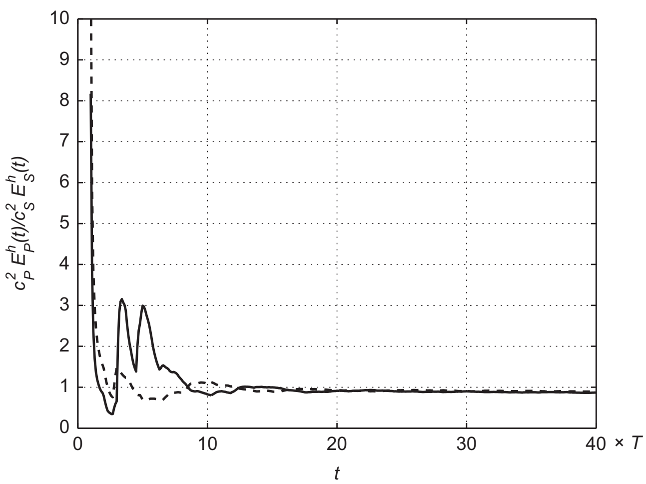

Fig. 10. Evolution of the ratios $c_{Pj}^2\mathcal{E}_{Pj}^h(t)/c_{Sj}^2\mathcal{E}_{Sj}^h(t)$, $j=1,2$, of the total bending/longitudinal and transverse energies in an assembly of two randomly heterogeneous cylindrical shells. Solid line: shell #1 ($j=1$), dashed line: shell #2 ($j=2$). After [93].

velocities c_S (subsystem #2) and c_P (subsystem #3), their respective mean vibrational energies $\mathbb{E}\{\mathcal{E}_{mr}(t)\}$ satisfy, for $r=1,2,3$

$$\eta_r\mathbb{E}\{\mathcal{E}_{mr}(t)\}+\sum_{s\neq r}(\eta_{rs}\mathbb{E}\{\mathcal{E}_{mr}(t)\}-\eta_{sr}\mathbb{E}\{\mathcal{E}_{ms}(t)\})=\frac{1}{\omega}\mathbb{E}\{\Pi_{\text{in},r}\}.$$

In this example power is injected in subsystem #1 solely, such that the above equations yield

$$\frac{\mathbb{E}\{\mathcal{E}_{m3}(t)\}}{\mathbb{E}\{\mathcal{E}_{m2}(t)\}}=\frac{n_3}{n_2}\left(\frac{\eta_{31}(\eta_2+\eta_{21}+\eta_{23})+\eta_{21}\eta_{32}}{\eta_{21}(\eta_3+\eta_{31}+\eta_{32})+\eta_{31}\eta_{23}}\right),$$

invoking the reciprocity relation $n_r\eta_{rs}=n_s\eta_{sr}$ for $r\neq s$. For weakly damped systems $\eta_r\ll\sum_{s\neq r}\eta_{rs}$, and this reduces to

$$\frac{\mathbb{E}\{\mathcal{E}_{m3}(t)\}}{\mathbb{E}\{\mathcal{E}_{m2}(t)\}}\simeq\frac{n_3}{n_2}=\frac{c_S^2}{c_P^2}$$

from Eq. (39). The above result is exactly the limit exhibited in Fig. 10. The same analyses and conclusions hold for beam assemblies as shown by the examples in [87,93,135]. It should be reminded, however, that these computations ignore the contribution of the glancing rays possibly trapped by the boundaries and the junction line, because a theoretical model lacks at present to describe them. Indeed it is known from experiments that the vibrational energy is very much likely to be guided by the interfaces, stiffeners or junctions in built-up structure, so these effects should not be disregarded. Further developments are needed on this issue.

5. Conclusions

Unsteady kinetic models for the vibrational energy evolution in (possibly random) heterogeneous slender structures have been outlined in this paper. The transport model equations can be solved numerically by Eulerian (discontinuous finite elements) and Lagrangian (Monte-Carlo) methods. A numerical example using the Monte-Carlo method for a shell assembly has been presented in order to illustrate the connection of the proposed theory with SEA. Other direct numerical simulations for different structural assemblies in [35,38,87,93,135] illustrate the transport regime for these systems, and the onset of a diffusive regime at late times. The latter is characterised by energy equilibration rules very much comparable to the assumption of modal equipartition invoked in the statistical energy analysis (SEA) of structural-acoustics systems [11]. The proposed approach is however believed to provide a rational theoretical framework for the validation and generalisation of heuristics such as SEA or the vibrational conductivity analogy (VCA). Both SEA and VCA are based on steady global or local diffusion equations established on some rather restrictive hypotheses.

The development of transport and diffusion kinetic models has been very active in the last twenty years, both in the applied mathematics and physics literature. Among the actual and future researches to be undertaken the following topics may be emphasised:

- extending the existing models to anisotropic media, such as composite structures, a task for which a robust theoretical framework has already been developed [26,34];
- extending the existing models to higher-order kinematics for slender structures, since they shall be more adapted to the high-frequency range than the first-order kinematics considered up to now;
- extending the existing models for the consideration of dissipative structural joints. The proposed theoretical approach is indeed clearly consistent with the experimental approach used to define losses at structural joints by equilibrating the power flows at interfaces.

Deriving boundary conditions for quadratic quantities such as the energy or power flow densities still constitutes an open, difficult issue. This holds especially true in the glancing set of energy rays – the diffractive and gliding ones – for the part of the power flows possibly trapped by the interfaces and junctions in built-up systems. This is critical for applications in structural dynamics, non-destructive evaluation techniques, or the development of theoretical and numerical models for multiphysics coupling. The different works undertaken in this direction shall benefit from the multimodel feature of the proposed approach in that it handles scalar energetic observables which are consistent from one physical domain to another. The analysis of approximate diffusion models for the transport regime, in particular the ones corresponding to multiple scattering at the boundaries and interfaces, shall be continued as well. The objectives are to identify characteristic transport and diffusion parameters for high-frequency structural dynamics, such as a diffusion coefficient or a mean free path related to the multiple scattering of waves at the boundaries and interfaces. These are typically the physical parameters considered in room acoustics, *e.g.* for the design of concert halls or sound insulation devices [117]. The advantages of diffusive models are twofold. First, diffusion equations are easier to solve numerically than transport equations. Second, they constitute a robust theoretical basis for the analysis of complex phenomena such as the weak and strong localisation of waves, the dynamic behaviour of heterogeneous materials, or the assessment of the range of validity of engineering methods such as SEA or VCA. It should be noted to conclude that the theoretical framework outlined above is not restricted to elastic waves although the latter were primarily investigated in this paper. All results apply to acoustic, electromagnetic or quantum waves as well. They shall be extended straightforwardly to applications in optics, underwater acoustics, or aeroacoustics for example.

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